Apply paint (painting)

The painting simulation: paint physics, tasks, assisted episodes and runtime

A US bedroom being repainted by the Franka Mobile FR3 Duo, with G1 standing craft support; paint is a conserved field on every surface (flow, levelling, sag, drying, absorption, laps), moved only by the roller nap, the brush tufts, the aerosol valve and spray, and drops.

Acceptance checks of the final physics: 6/9 pass.

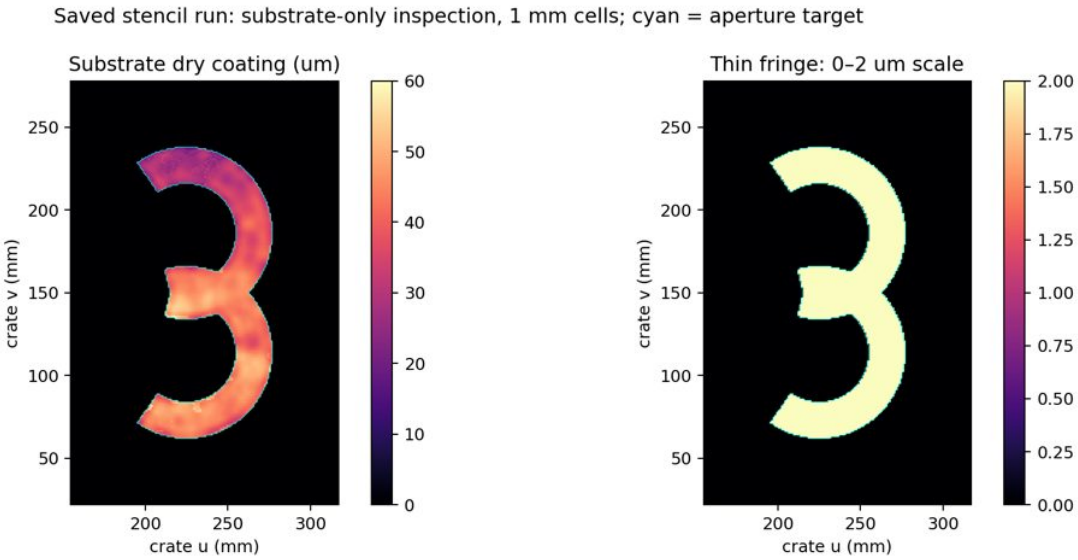
check	criterion	value	result
conserved overspray accounting	wet/dry invariant; both spray judgments on nonvolatile solids at the 5% limit	{"spray-stencil": {"fraction": 0.046018, "basis": "nonvolatile solids volume", "outside_so	PASS
local contact seal	flush edge blocked; all tested gaps >= 0.05 mm permit bleed; conserved solids	[{"gap_mm": 0, "blocked_cells": 113448, "under_edge_solids_fraction": 0.0, "input_solids_m	PASS
direct corner component	no artificial bead; corner hidden >= 0.90 and ceiling paint <= 0.15 cm2 over the 0.30 m component	{"corner_hidden_frac": 0.255, "ceiling_cm2_whole_row": 0.0, "top_gap_mm_median": 2.0}	FAIL
bounded optical estimate	three colour pairs tested at low/nominal/high class estimates; no product hide claim	[{"color": "lullaby", "background": "naval", "white_base_S_per_um": 0.08, "hide_dft_um": 7	PASS
fine aerosol appearance	mist and deposited fringe not visibly parcel-sized at working distance	Smaller plume markers and a continuous stencil substrate field, but opaque surface-sink dots and low-density fringe remain visibly coarse. Can coverage is an unvalidated calibration uncertainty.	FAIL
representative grasp event	time-aligned contacts and onset movie; refined rigid replay retains actual finger contact, no fault	{"max_slip_mm": 26.154828632275755, "final_slip_mm": 8.733874985442364, "final_finger_norm	PASS
feathered touch-up of a scuff	brush-mediated scuff contrast <= 1 and ridge <= 25 um, task evaluator	{"scuffs_hidden": {"value": 1.967, "limit": 1.0, "ok": false}, "no_ridge": {"value": 92.23	FAIL
renderer lifecycle	cold action executes 0 ticks; explicit timed readiness; next 40-tick RGB HTTP action <= 200x with trace	{"readiness_wall_s": 11.611859918572009, "first_action_ratio": 25.84775947034359, "trace_f	PASS
G1 craft interaction	shared humanoid rig, 60 s rest, public physical brush stroke/lift, RGB, proprioception and recording	{"component_supported": true, "film_solids_mL": 0.005823152855230126, "maximum_grip_slip_m	PASS

Not claimed: Autonomous task solutions (assisted episodes use a phantom operator; robot demos are scripted); Measured-product calibration (parameters rest on published data sheets, standards and estimates); A per-bristle brush or a resolved spray breakup (pseudo-rigid tufts; a calibrated parcel jet); A resolved meniscus at inside corners (the unsupported bead shortcut is disabled) or contact-patch geometry for the robot's paint skin (a fixed 4 mm footprint); Robot configurations beyond the qualified Mobile FR3 variants and G1 craft component; Real-time simulation

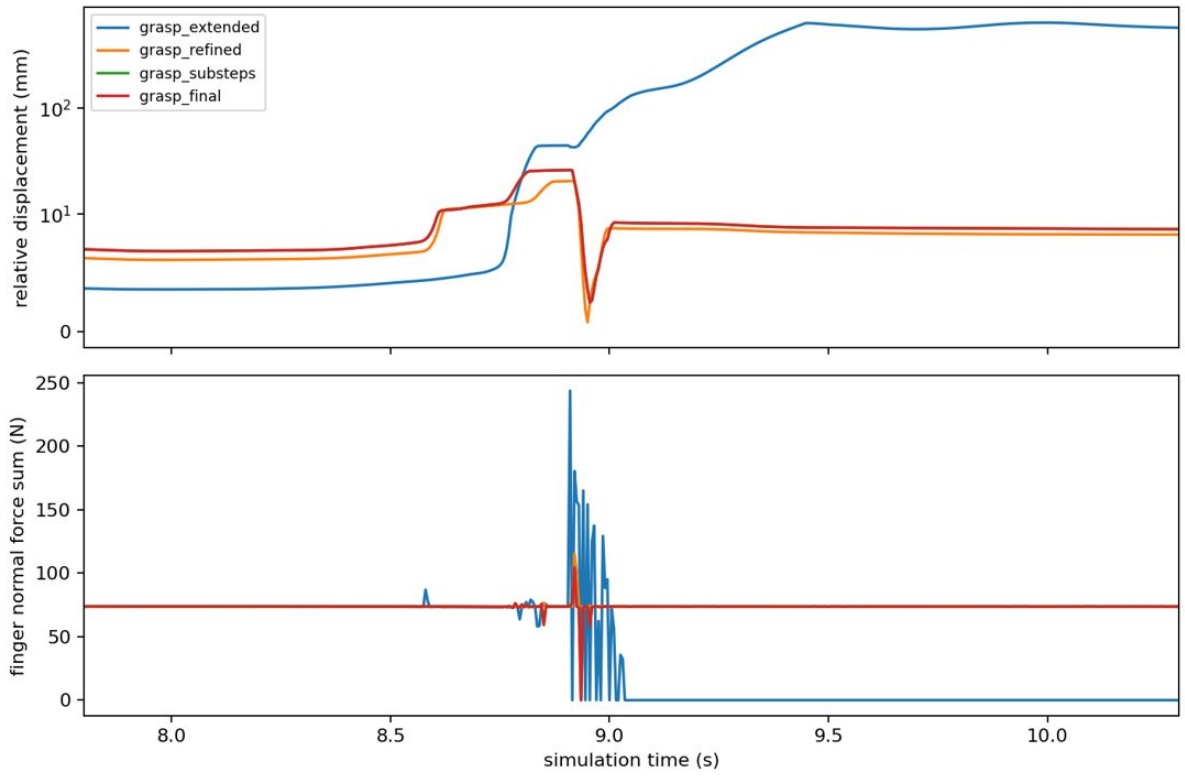
These checks cover separate physical components; no full task solve or autonomous policy is required. Sections 1 to 5 report measurements made with an earlier version of the physics (0.5 ms rigid steps, an earlier overspray accounting and a one-face corner bead transfer). They are not reruns with the final solver, and three of their results do not hold under it: the clear-coat episode's containment pass, the corner hiding in the cut-line study, and the reading of the P18 torque comparison as a grip overload.

Final physics: containment, stencil seal, optics, grasp and G1 craft

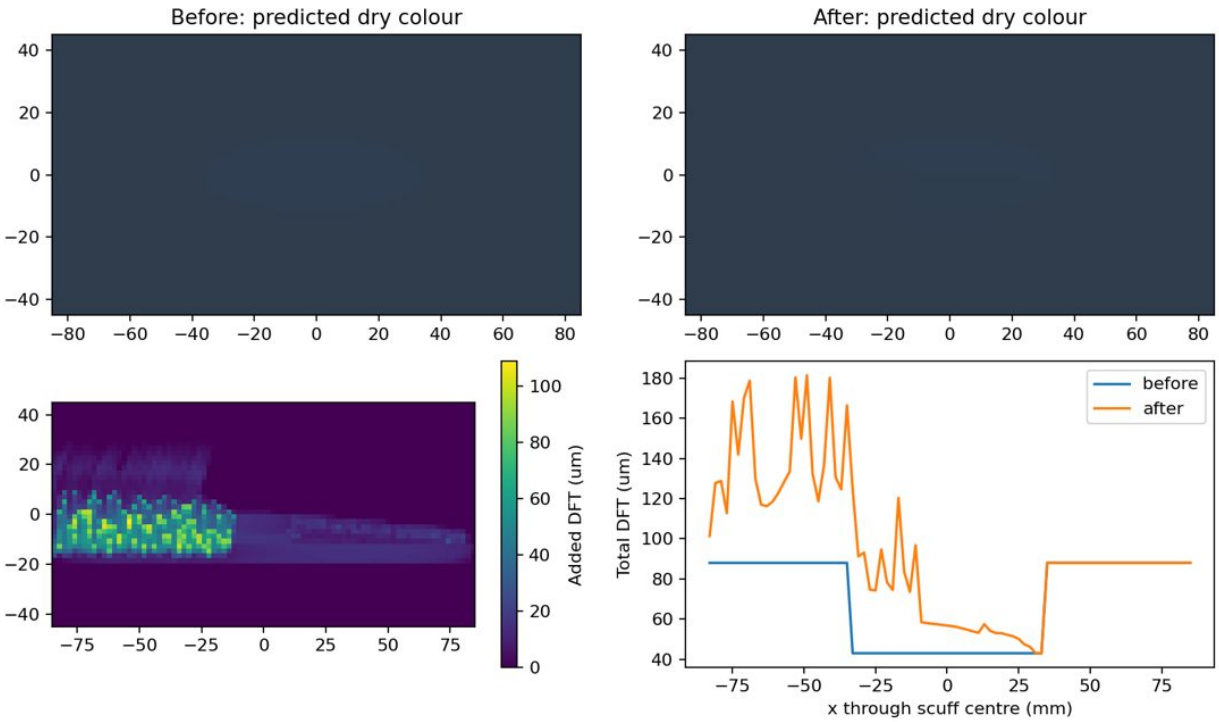
Containment compares conserved nonvolatile solids, including outside sinks and airborne material. The stencil seals only within a local 20 um ideal contact tolerance; open gaps do not seal. The white optical coefficient is an estimate, and its coating-class sensitivity range changes nominal one- or two-coat judgments. There is no corner bead transfer: paint reaches a protected face only through brush contact. The rigid world steps in two 0.25 ms substeps per 0.5 ms control tick.



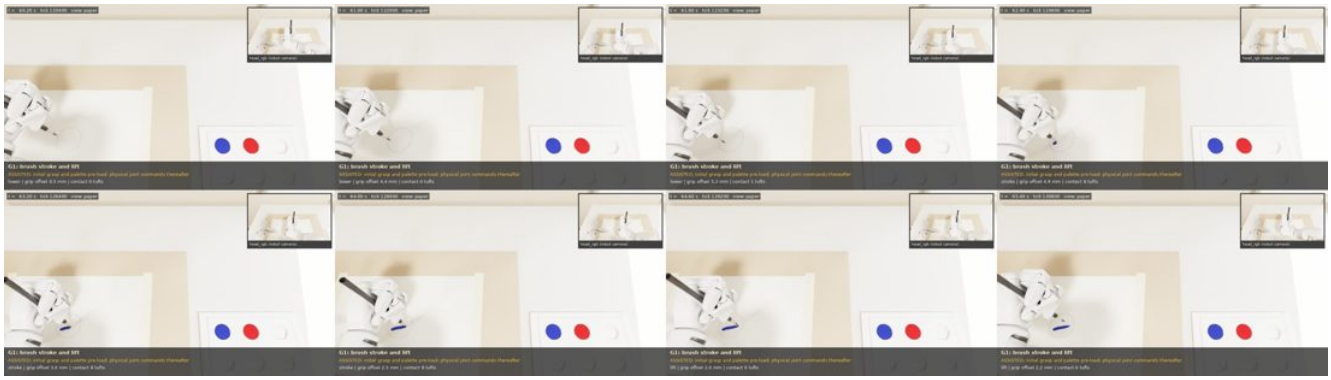
Stencil substrate only, after the spray: actual DFT, no smoothing, 1 mm cells.



Time-aligned can motion: the loss of the can depends on the rigid timestep. With 0.5 ms rigid steps the can is lost; with two 0.25 ms rigid substeps per tick, as the environment uses, the fingers keep contact. An upper torque bound does not show a physical overload.



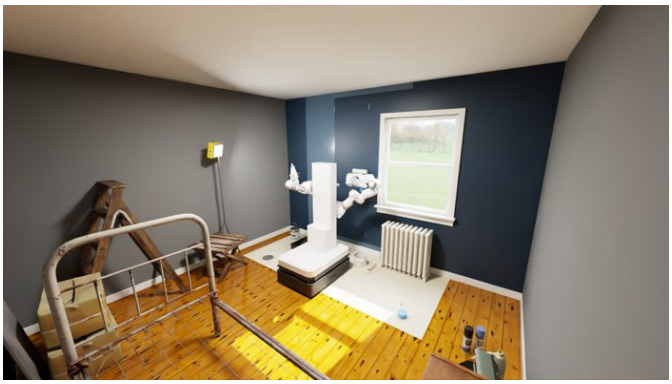
One-scuff brush touch-up: measured local colour and thickness. The contrast and ridge criteria fail.



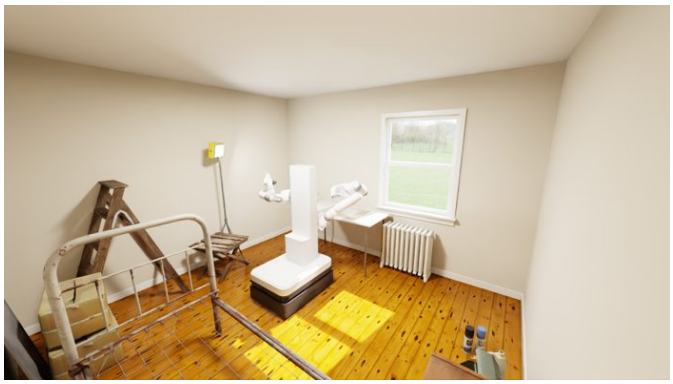
G1 standing craft component: assisted initial grasp/preload, physical joint commands and brush contact. No pelvis support or grasp weld.

Sections 1 to 5 below come from the earlier physics version described on the first page and are not reruns with the final solver.

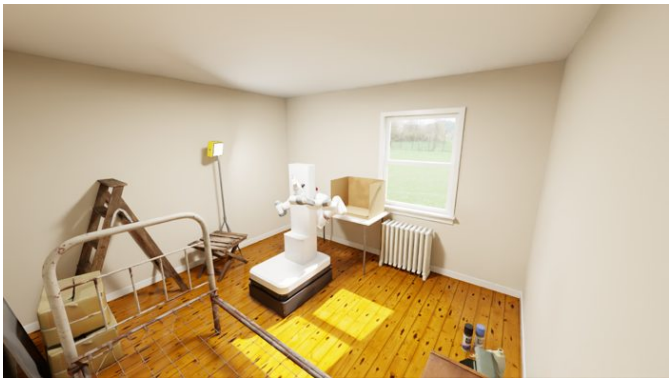
1. Scene: every task's start state



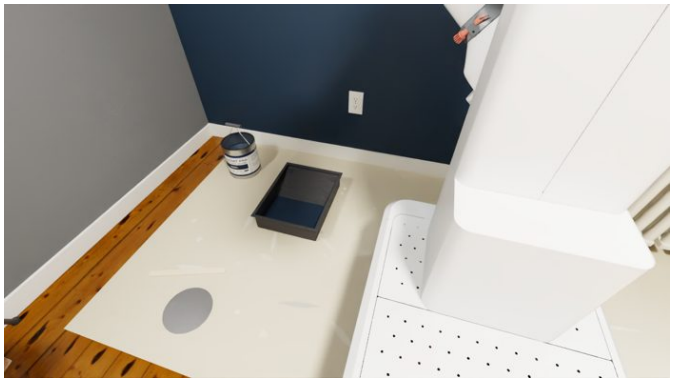
the room (wall preset)



craft preset



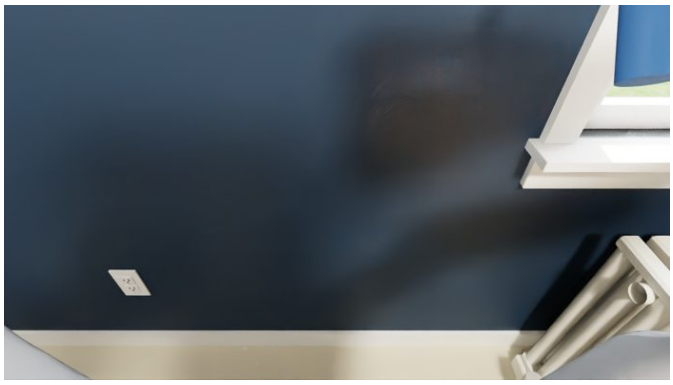
spray preset



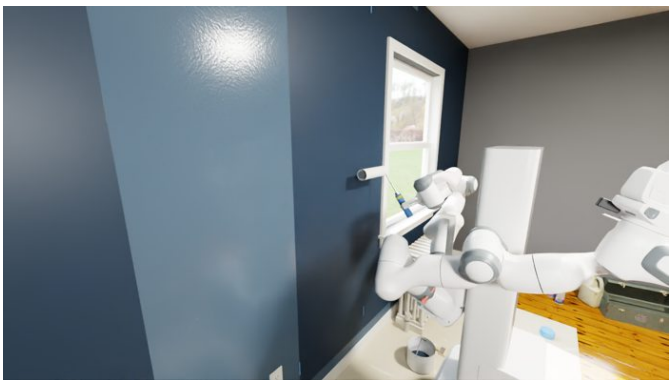
tools on the cloth



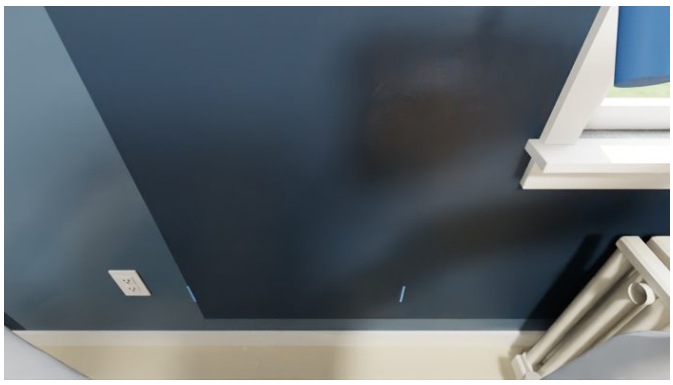
load-roller: tray_close (diagnostic)



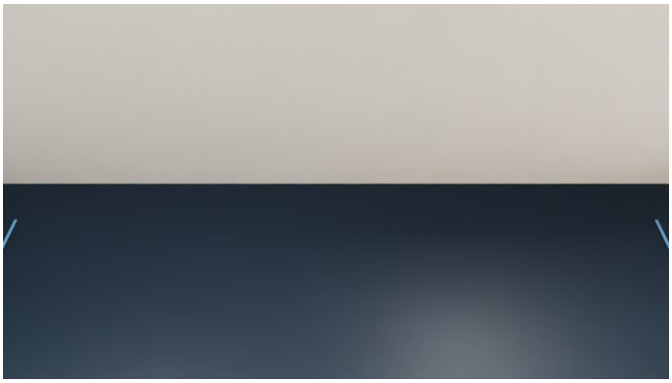
load-roller: head_rgb (the robot's camera)



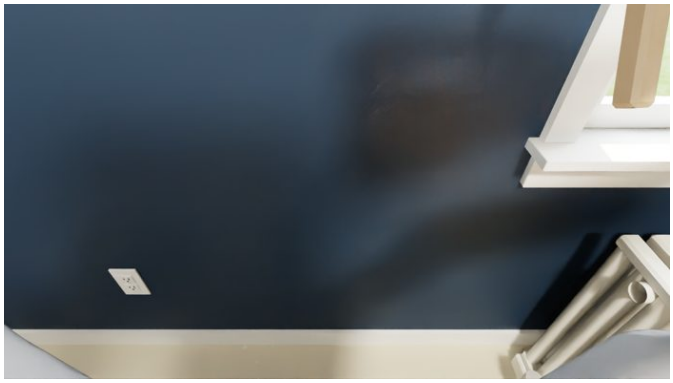
roll-strip: strip (diagnostic)



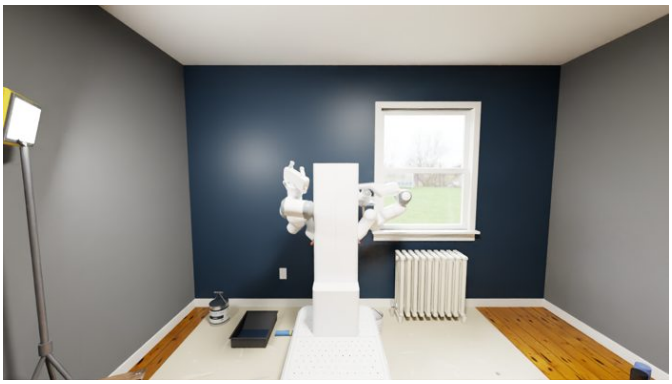
roll-strip: head_rgb (the robot's camera)



cut-in-ceiling: ceiling_line (diagnostic)



cut-in-ceiling: head_rgb (the robot's camera)



cut-in-trim: wall (diagnostic)



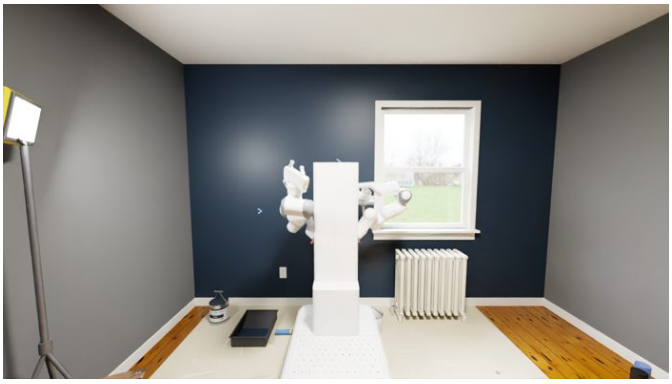
cut-in-trim: head_rgb (the robot's camera)



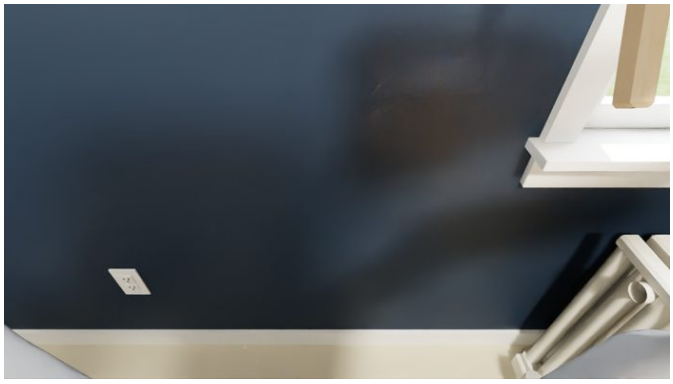
behind-radiator: radiator_end (diagnostic)



behind-radiator: head_rgb (the robot's camera)



touch-up: wall (diagnostic)



touch-up: head_rgb (the robot's camera)



fill-circle: paper (diagnostic)



fill-circle: head_rgb (the robot's camera)



brush-stroke: paper (diagnostic)



brush-stroke: head_rgb (the robot's camera)



swatch-card: paper (diagnostic)



swatch-card: head_rgb (the robot's camera)



spray-stencil: stencil (diagnostic)



spray-stencil: head_rgb (the robot's camera)

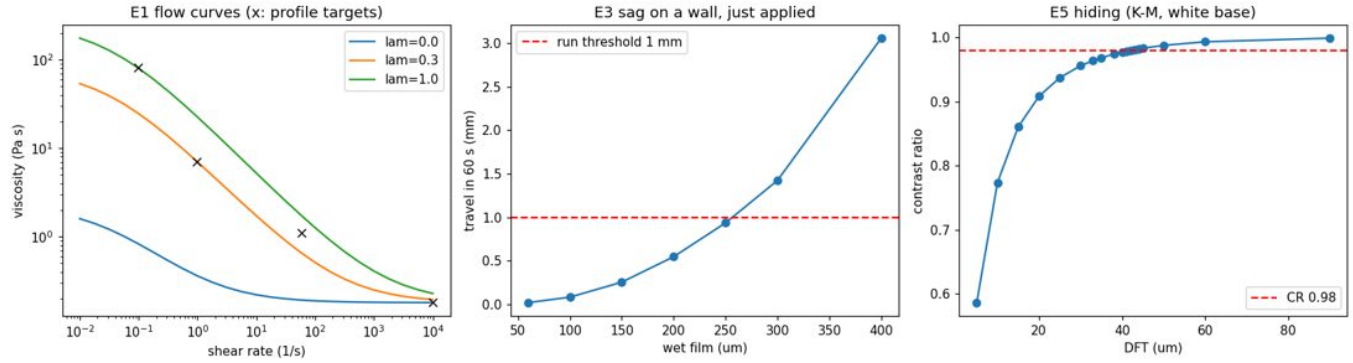


clear-coat: spray_box (diagnostic)

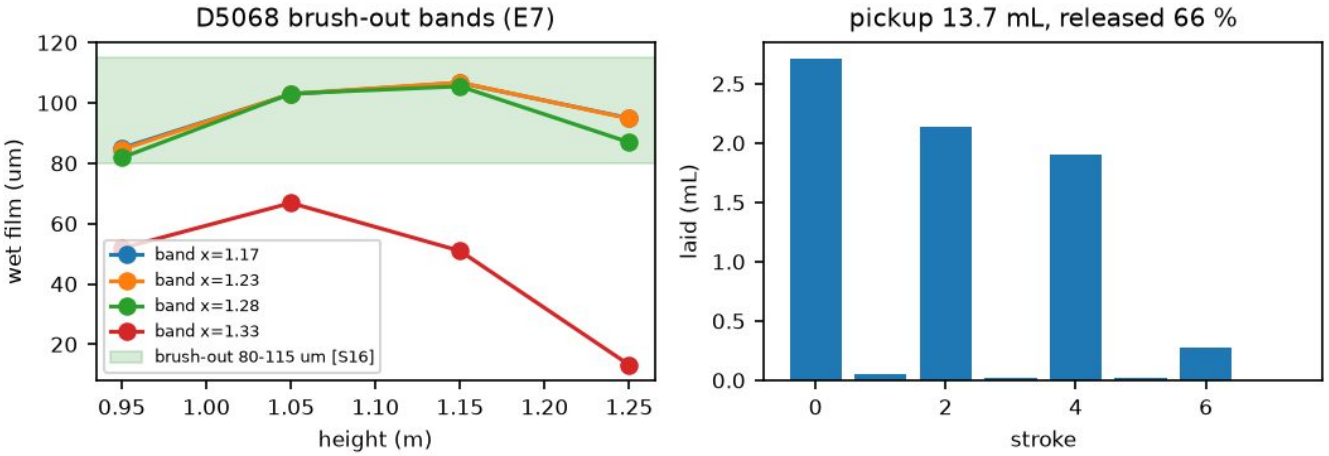


clear-coat: head_rgb (the robot's camera)

2. Component physics (assisted benches)



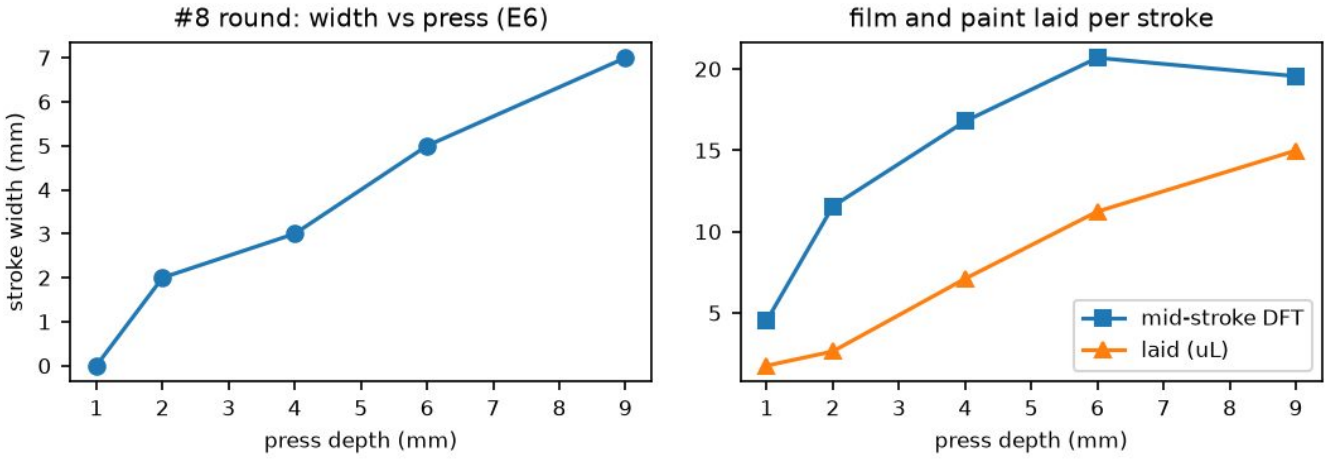
P2 film benches: rheology, levelling vs Orchard, sag, wet edge, hiding, paper absorption



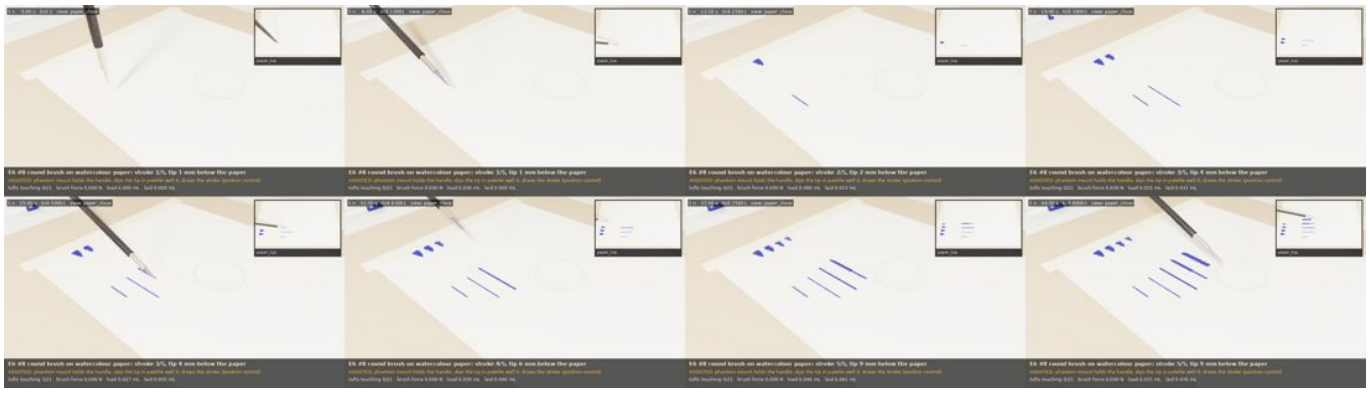
E7 sash brush ASTM D5068 brush-out on a phantom mount (pickup, release, band thickness)



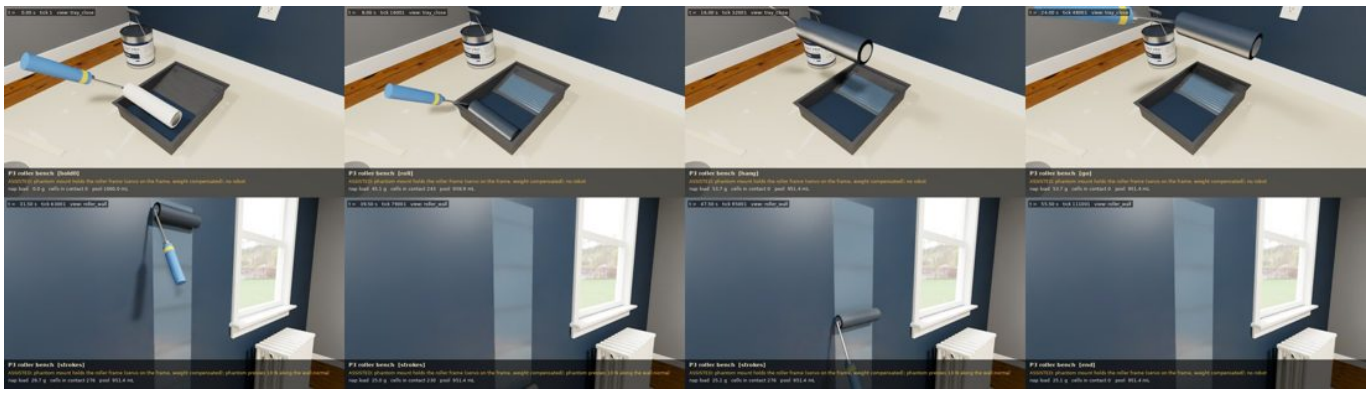
E7 video contact sheet



E6 #8 round on paper: press depth controls width (e-David trend)



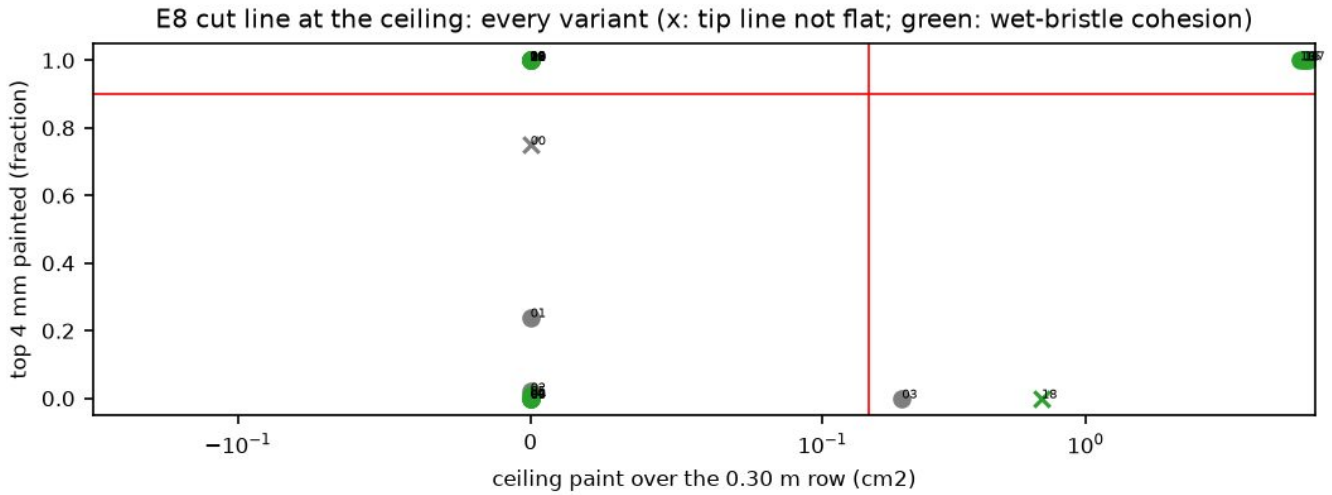
E6 video contact sheet



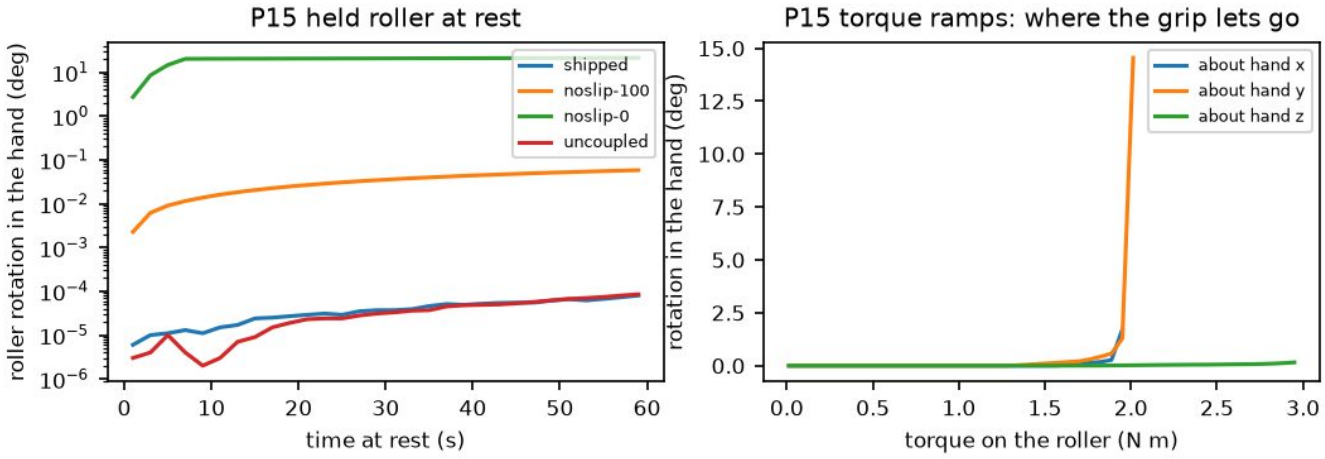
P3 roller: tray loading, drip hold, rolling at 10 N (contact sheet)



P5 aerosol: phantom press on the valve, spray through the stencil (contact sheet)



E8 cut-line reachability: ceiling paint against the corner strip painted, every holding variant



P15 a held tool at rest (no-slip convergence, coupled fingers) and the grip's torque capacity

phase	force (N)	button (mm)	open
rest after the reset settle	0.0	-0.001	0.0
10 N on the button along the can axis, 1 s	10.0	0.004	0.0
released after 10 N, 0.5 s	0.0	-0.001	0.0
25 N on the button along the can axis, 1 s	25.0	2.004	1.0
released after 25 N, 0.5 s	0.0	-0.001	0.0
45 N on the button along the can axis, 1 s	45.0	2.202	1.0
released after 45 N, 0.5 s	0.0	-0.001	0.0
80 N on the button along the can axis, 1 s	80.0	2.205	1.0
released after 80 N, 0.5 s	0.0	-0.001	0.0
120 N on the button along the can axis, 1 s	120.0	2.207	1.0
released after 120 N, 0.5 s	0.0	-0.001	0.0

P16 aerosol button against its 0-2.2 mm slide (pure force along the can axis).

case	same verdict	fast wall (s)	full wall (s)	settle (s)
roll-start	True	2.3	65.7	546.0
spray-coat	True	0.6	3.4	140.0
roll-lane	True	8.4	63.9	1620.0

P17 settle rule: fast path (stopped films only dried) against the full settle.

aim above the scuff (mm)	mark centre (mm)	laid (mL)	new film on the scuff p10 / median / max (um)
-20.0	-32.3	0.023	0.0 / 0.0 / 0.0
-10.0	-22.3	0.023	0.0 / 0.0 / 8.1
0.0	-12.3	0.023	0.0 / 0.9 / 14.3
10.0	-2.3	0.023	0.0 / 6.0 / 14.3
15.0	2.6	0.023	0.0 / 5.6 / 14.4
20.0	7.7	0.023	0.0 / 4.2 / 13.9
25.0	12.5	0.023	0.0 / 2.5 / 9.8
30.0	17.7	0.023	0.0 / 1.0 / 9.8

E9 touch-up aim: where one light stroke of the sash brush lands against where its tip line is aimed (phantom mount).

row z (m)	from	head off wall (mm)	laid (mL)	half row reached	new film (um)	hidden	radiator (mL)	wet wall touched (cm2)
0.183	left	3.0	1.95	1.0	39.2	1.0	0	0.0
0.396	left	3.0	1.98	1.0	39.5	1.0	0	0.0
0.678	left	3.0	2.03	1.0	39.3	1.0	0	0.0
0.396	right	3.0	1.91	1.0	38.8	1.0	0	0.0
0.396	left	6.0	1.98	1.0	39.5	1.0	0	0.0

row z (m)	from	head off wall (mm)	laid (mL)	half row reached	new film (um)	hidden	radiator (mL)	wet wall touched (cm2)
0.118	left	3.0	1.96	1.0	39.3	1.0	0	0.0

E10 behind the radiator: one row of the long-reach brush pushed into the 64 mm gap from an end (phantom mount).

task	tool	hand peak (m/s, rad/s)	torque asked (N m)	max in hand (mm)	left the hand
load-roller	roller	1.57, 7.59	16.28	1550.47	P11 random motion #1 (joints)
cut-in-ceiling	sash	1.54, 7.59	1.631	100.01	wrist sweep 3: joints 5-7 to 10 / 90 % of their range in 0.4
behind-radiator	longreach	1.55, 7.59	6.325	253.5	wrist sweep 1: joints 5-7 to 10 / 90 % of their range in 0.4
fill-circle	round8	3.04, 9.92	0.1	4.58	no
spray-stencil	aerosol	3.48, 10.44	0.954	576.39	return to the start pose

P18 start grasps under fast legal arm motion (earlier physics, 0.5 ms rigid steps). The torque is an upper estimate, not the actual required torque, and its comparison with the minimum static capacity does not establish a physical cause of loss. The time-aligned replay on the first pages shows that the loss depends on the rigid timestep.

3. Assisted task episodes through the evaluator

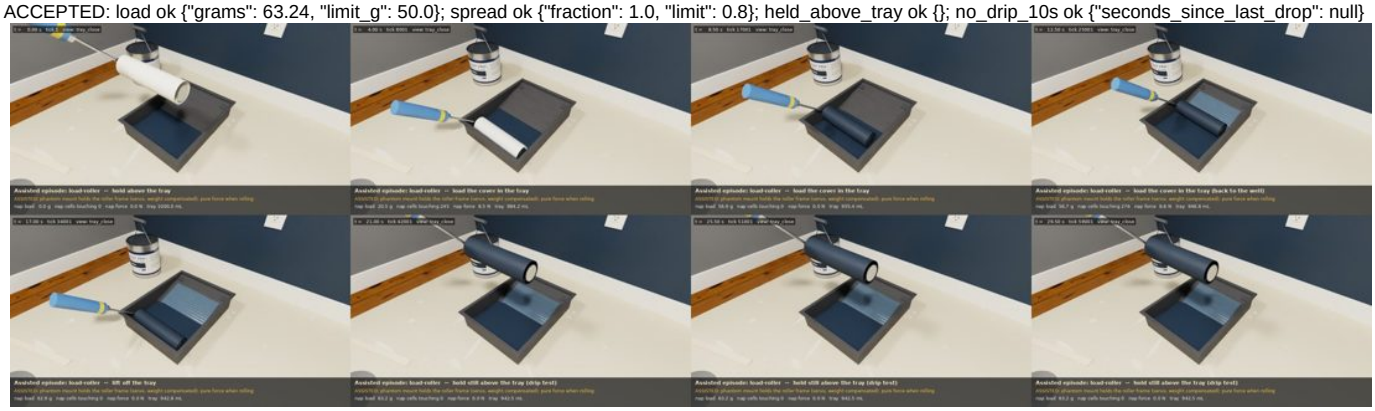
A phantom operator holds the tool where a hand would (six-axis servo, weight compensated), presses with a pure force along the surface normal or presses the aerosol button; nothing else touches the world. The shipped evaluator judges the result with its settle rule. Film maps are computed from the film fields.

task	accepted	failed criteria	sim (s)	wall (s)
load-roller	True	-	29.72	485.2
roll-strip	True	-	175.18	4741.3
cut-in-ceiling	False	hide	256.57	6961.6
cut-in-trim	False	hide, no_laps, protected	569.5	11091.6
behind-radiator	True	-	523.7	15244.9
touch-up	False	scuffs_hidden, no_ridge	163.65	4170.2
fill-circle	False	coverage	260.39	2921.0
brush-stroke	False	coverage	87.53	1980.1
swatch-card	False	coverage	215.8	2715.1
spray-stencil	True	-	95.3	1330.2
clear-coat	True	-	99.7	1899.5

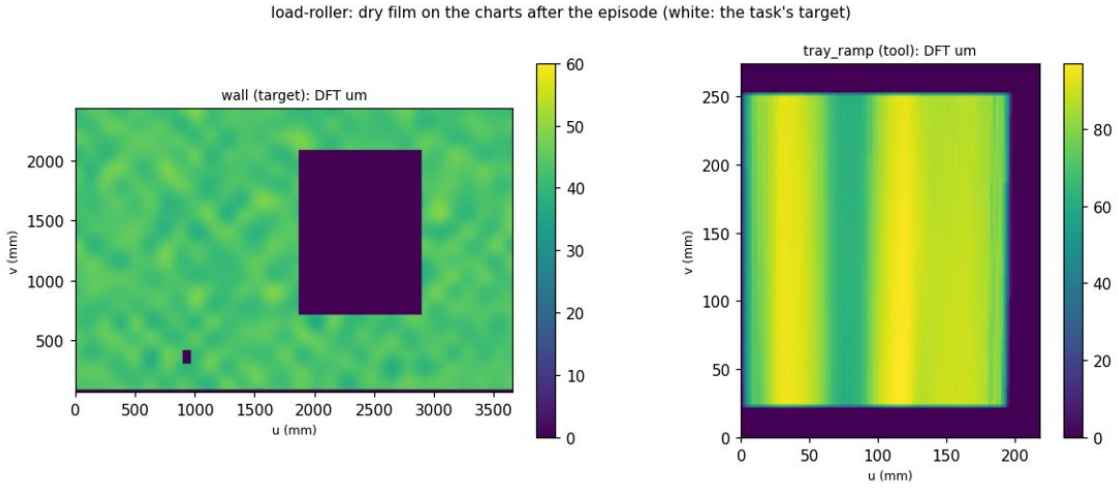
Under the final overspray accounting (nonvolatile solids outside the target over solids emitted), a replay of the clear-coat episode puts 20.17 % outside against the 5 % limit and fails containment; its other checks pass. The stencil episode's replay puts 4.60 % outside and passes.

load-roller

ACCEPTED: load ok {"grams": 63.24, "limit_g": 50.0}; spread ok {"fraction": 1.0, "limit": 0.8}; held_above_tray ok {}; no_drip_10s ok {"seconds_since_last_drop": null}



load-roller: video contact sheet



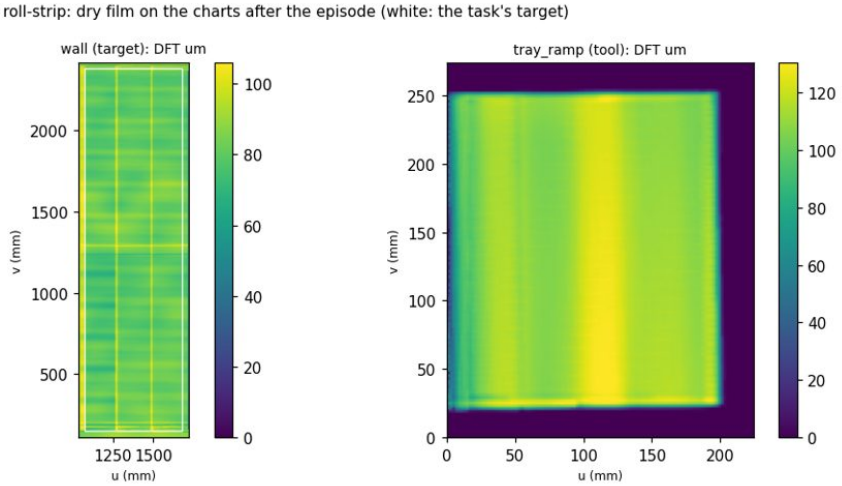
load-roller: dry film after the episode (white outline: the task's target)

roll-strip

ACCEPTED: hide ok {"value": 0.9996, "limit": 0.98, "dE_limit": 1.0, "dft_um_p5_median": [74.8, 80.7]}; no_runs ok {"wet_max_um": 0.0, "travel_mm": 0.014}; no_laps ok {"area_cm2": 0.88, "limit_cm2": 1.0}; uniform ok {"dE": 0.261, "limit": 1.0, "note": null}; protected ok {"area_cm2": 0.0, "limit_cm2": 0.5}; untouched ok {"area_cm2": 0.0, "limit_cm2": 0.5}; roller_in_tray ok {"cover_in_tray": true, "released": true}



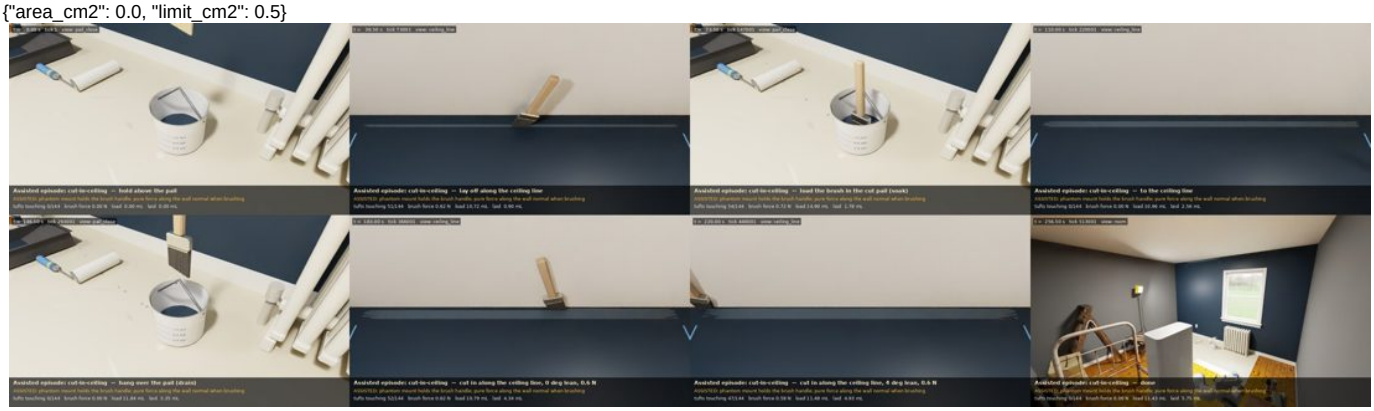
roll-strip: video contact sheet



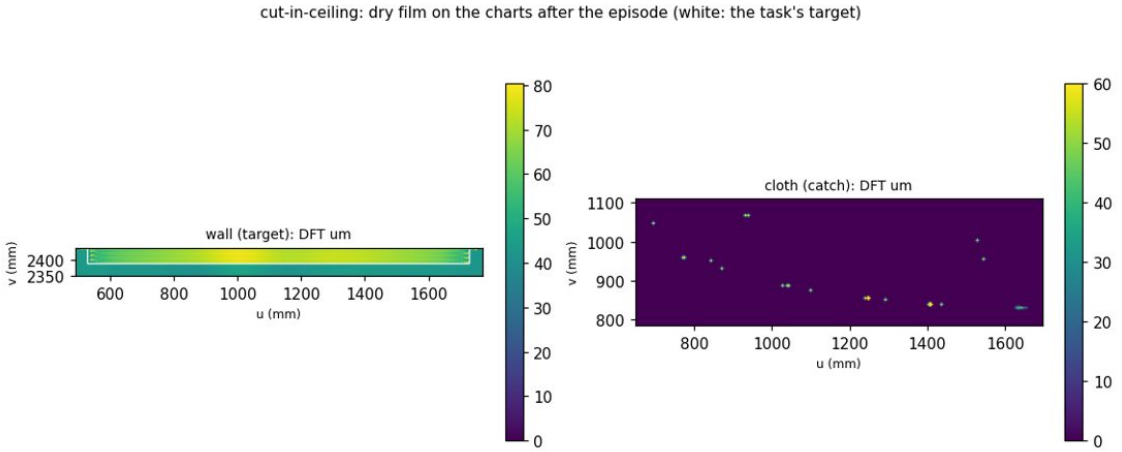
roll-strip: dry film after the episode (white outline: the task's target)

cut-in-ceiling

NOT ACCEPTED: hide FAIL {"value": 0.8495, "limit": 0.95, "dE_limit": 1.0, "dft_um_p5_median": [55.6, 70.5]}; no_runs ok {"wet_max_um": 0.0, "travel_mm": 0.016}; no_laps ok {"area_cm2": 0.0, "limit_cm2": 1.0}; reaches_corner ok {"value": 0.9217, "limit": 0.9}; protected ok {"area_cm2": 0.0, "limit_cm2": 0.5}; untouched ok {"area_cm2": 0.0, "limit_cm2": 0.5}



cut-in-ceiling: video contact sheet



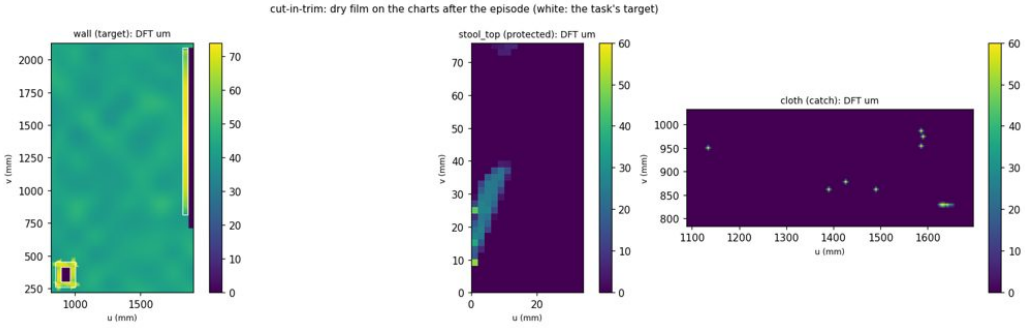
cut-in-ceiling: dry film after the episode (white outline: the task's target)

cut-in-trim

NOT ACCEPTED: hide FAIL {"value": 0.8162, "limit": 0.95, "dE_limit": 1.0, "dft_um_p5_median": [51.0, 69.8]}; no_runs ok {"wet_max_um": 0.0, "travel_mm": 0.056}; no_laps FAIL {"area_cm2": 2.16, "limit_cm2": 1.0}; protected FAIL {"area_cm2": 2.16, "limit_cm2": 0.5}; untouched ok {"area_cm2": 0.0, "limit_cm2": 0.5}



cut-in-trim: video contact sheet



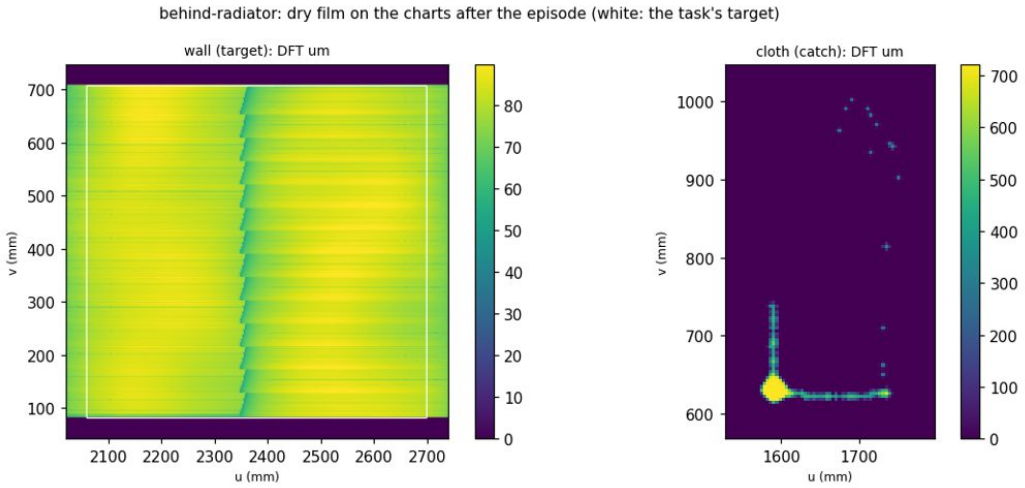
cut-in-trim: dry film after the episode (white outline: the task's target)

behind-radiator

ACCEPTED: hide ok {"value": 0.9922, "limit": 0.9, "dE_limit": 1.0, "dft_um_p5_median": [72.9, 83.1]}; no_runs ok {"wet_max_um": 0.0, "travel_mm": 0.002}; no_laps ok {"area_cm2": 0.12, "limit_cm2": 1.0}; protected ok {"area_cm2": 0.0, "limit_cm2": 0.5}; untouched ok {"area_cm2": 0.0, "limit_cm2": 0.5}



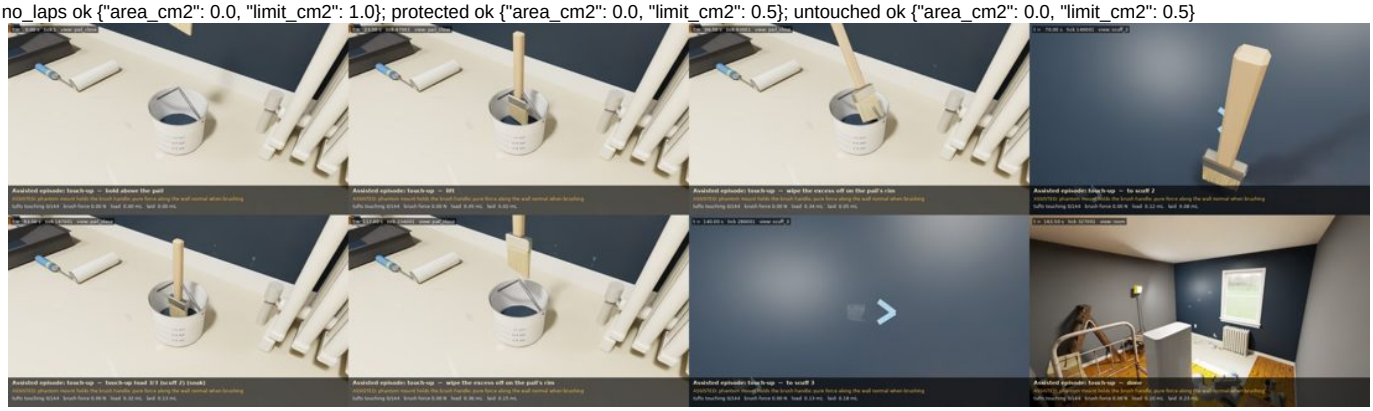
behind-radiator: video contact sheet



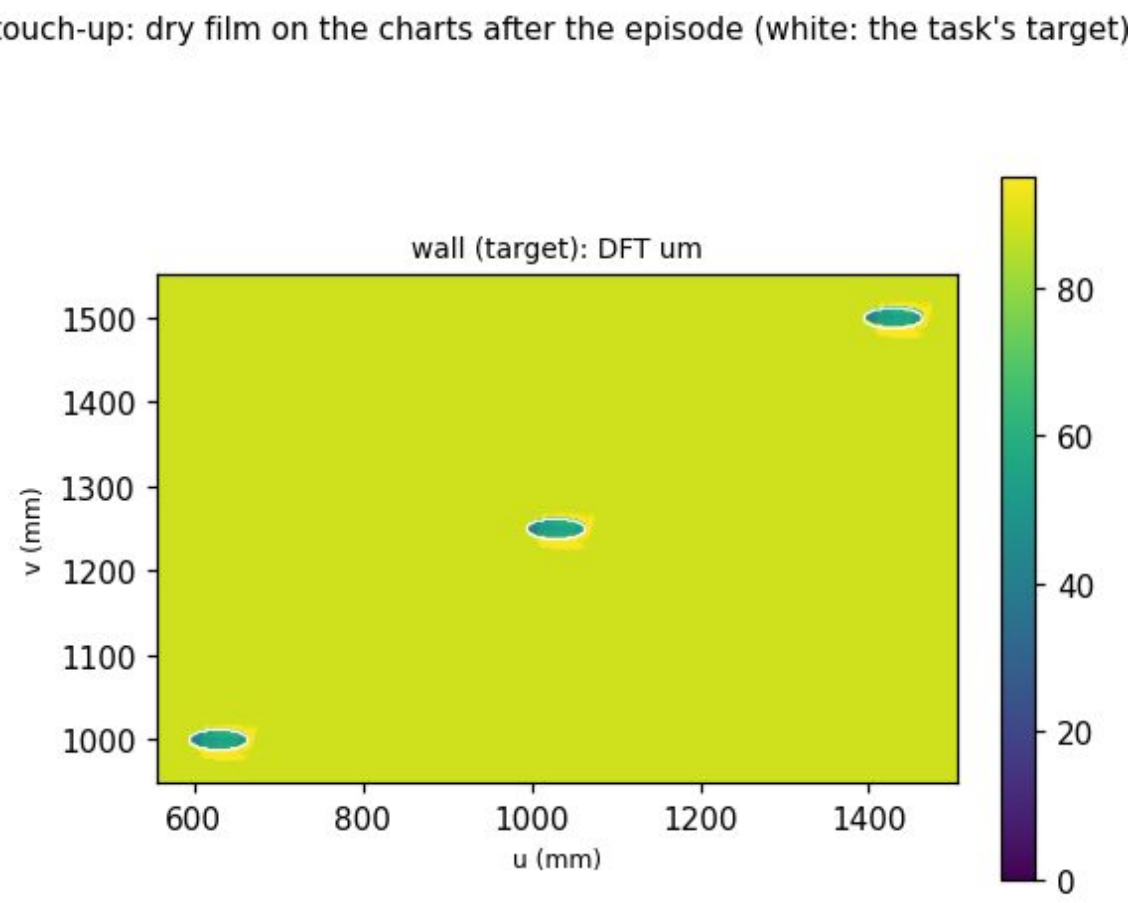
behind-radiator: dry film after the episode (white outline: the task's target)

touch-up

NOT ACCEPTED: scuffs_hidden FAIL {"value": 1.872, "limit": 1.0}; no_ridge FAIL {"value": 33.85, "limit_um": 25.0}; no_runs ok {"wet_max_um": 0.0, "travel_mm": 0.0}; no_laps ok {"area_cm2": 0.0, "limit_cm2": 1.0}; protected ok {"area_cm2": 0.0, "limit_cm2": 0.5}; untouched ok {"area_cm2": 0.0, "limit_cm2": 0.5}



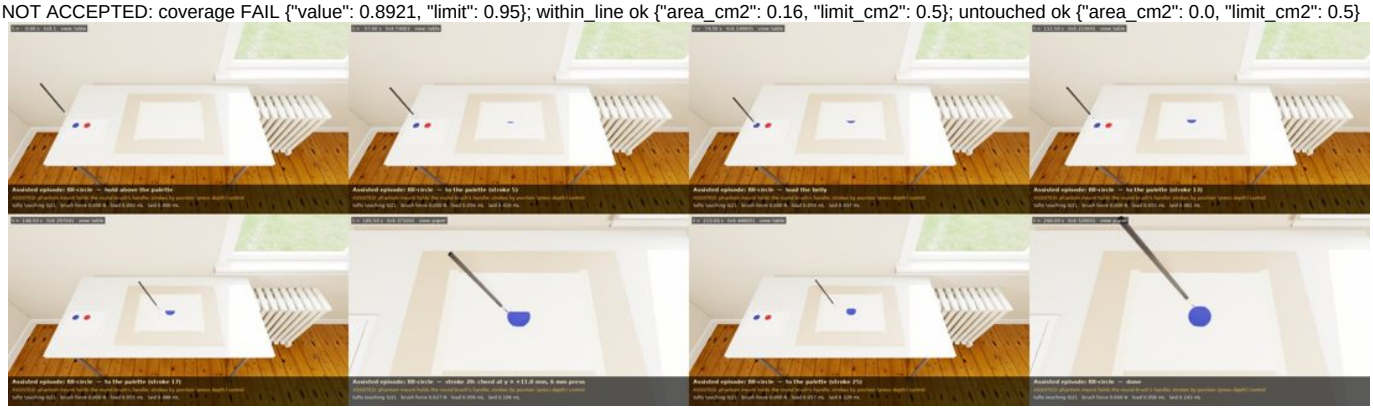
touch-up: video contact sheet



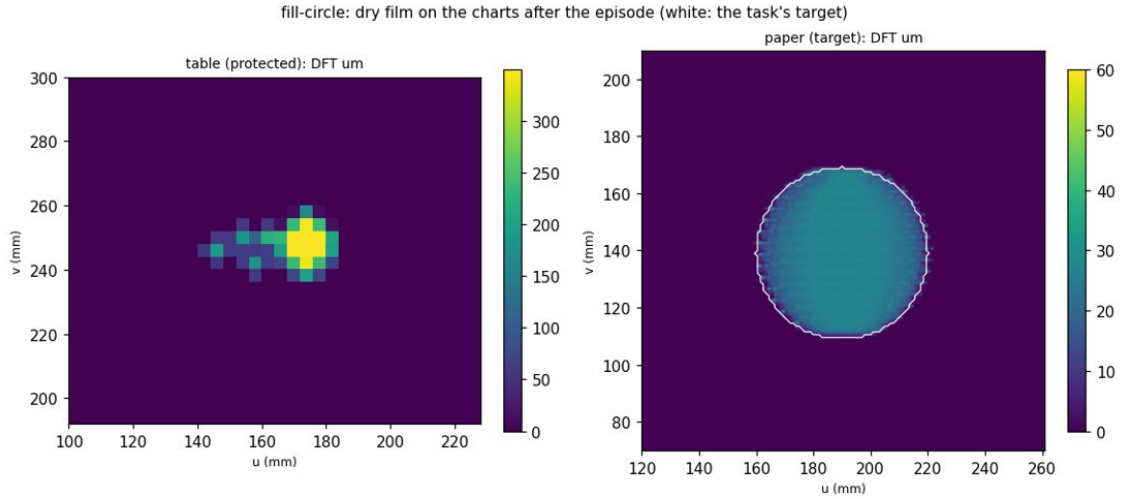
touch-up: dry film after the episode (white outline: the task's target)

fill-circle

NOT ACCEPTED: coverage FAIL {"value": 0.8921, "limit": 0.95}; within_line ok {"area_cm2": 0.16, "limit_cm2": 0.5}; untouched ok {"area_cm2": 0.0, "limit_cm2": 0.5}



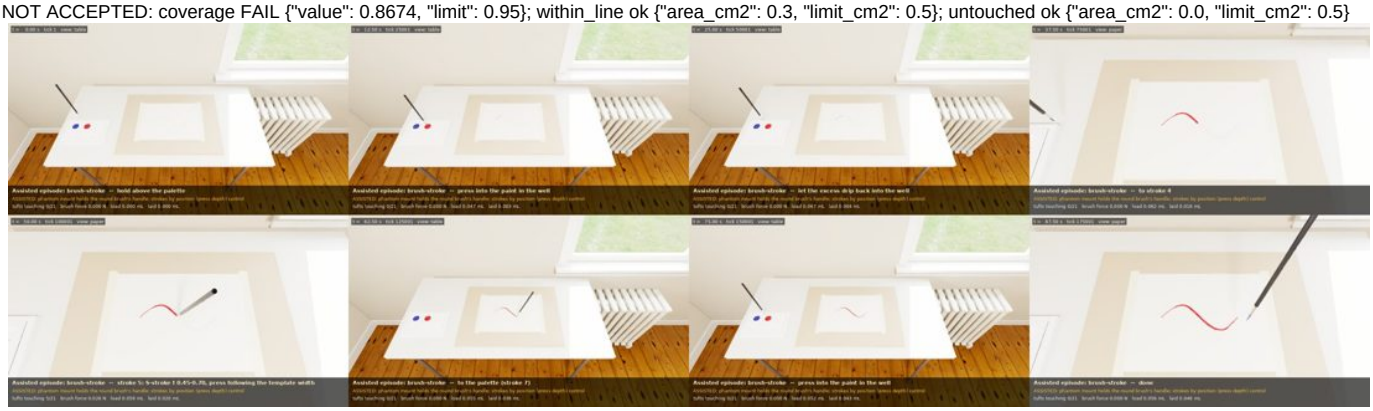
fill-circle: video contact sheet



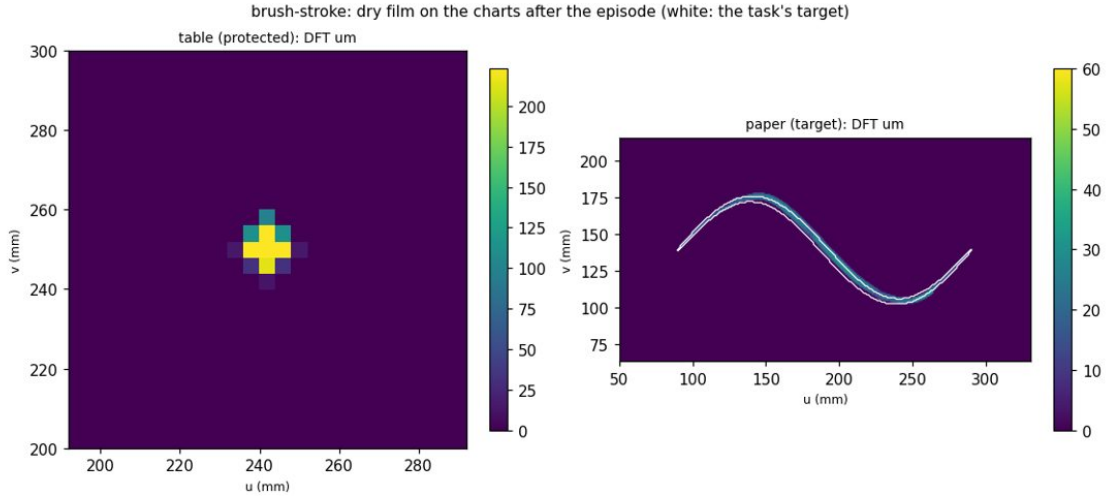
fill-circle: dry film after the episode (white outline: the task's target)

brush-stroke

NOT ACCEPTED: coverage FAIL {"value": 0.8674, "limit": 0.95}; within_line ok {"area_cm2": 0.3, "limit_cm2": 0.5}; untouched ok {"area_cm2": 0.0, "limit_cm2": 0.5}



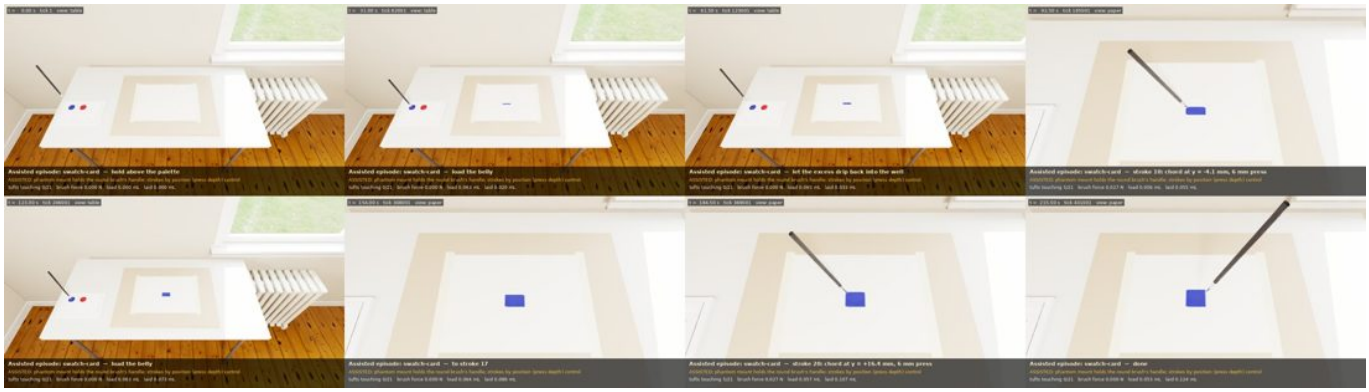
brush-stroke: video contact sheet



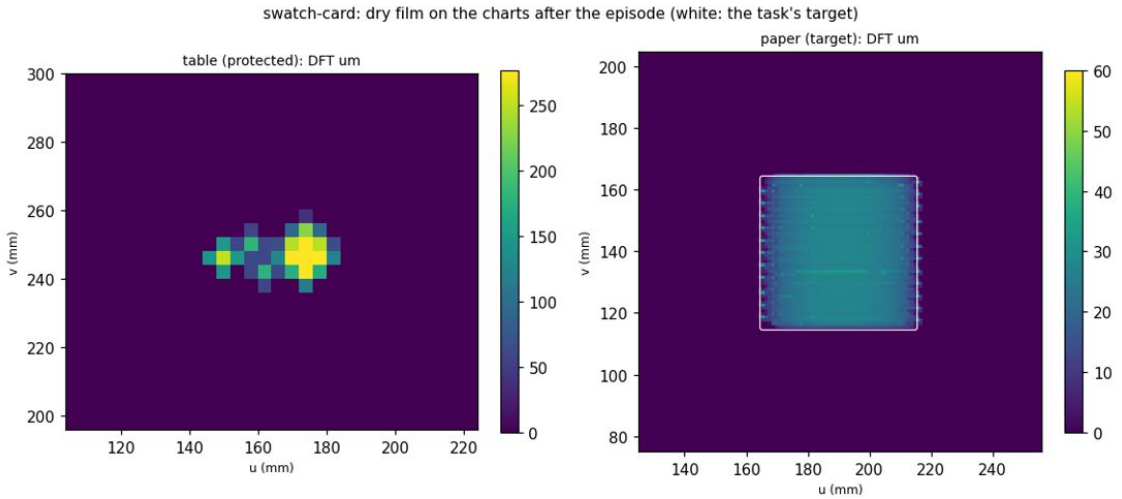
brush-stroke: dry film after the episode (white outline: the task's target)

swatch-card

NOT ACCEPTED: coverage FAIL {"value": 0.8682, "limit": 0.95}; within_line ok {"area_cm2": 0.0, "limit_cm2": 0.5}; untouched ok {"area_cm2": 0.0, "limit_cm2": 0.5}



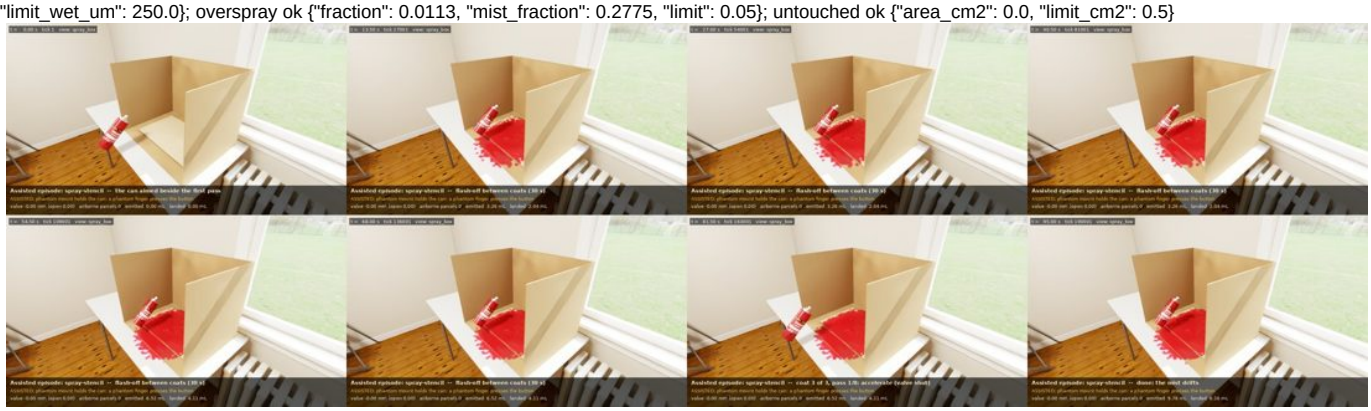
swatch-card: video contact sheet



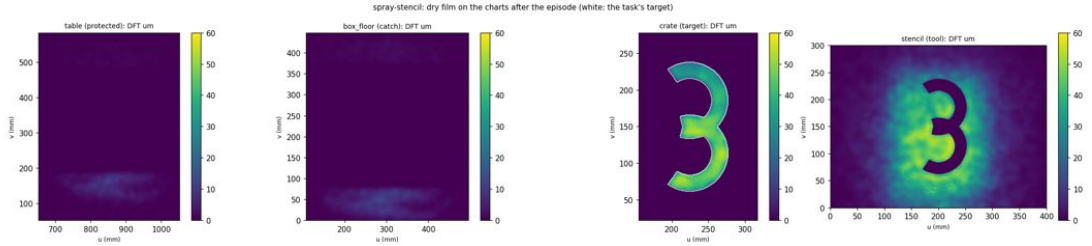
swatch-card: dry film after the episode (white outline: the task's target)

spray-stencil

ACCEPTED: coat ok {"value": 0.9521, "limit": 0.9}; sharp_edges ok {"outside_fraction": 0.0009, "limit": 0.02}; no_runs ok {"wet_max_um": 158.1, "travel_mm": 0.517, "limit_wet_um": 250.0}; overspray ok {"fraction": 0.0113, "mist_fraction": 0.2775, "limit": 0.05}; untouched ok {"area_cm2": 0.0, "limit_cm2": 0.5}



spray-stencil: video contact sheet



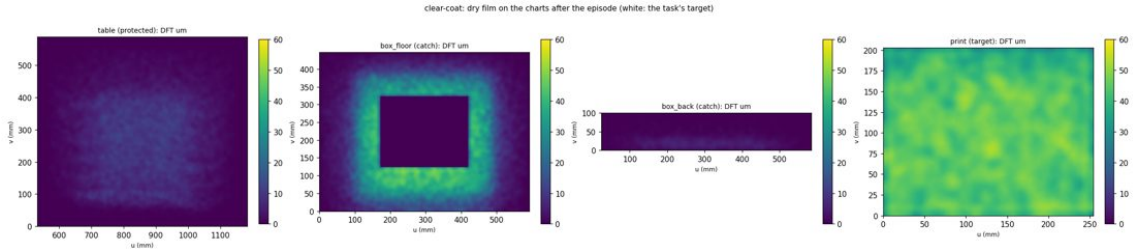
spray-stencil: dry film after the episode (white outline: the task's target)

clear-coat

ACCEPTED: coat ok {"value": 1.0, "limit": 0.95}; even ok {"max_over_min": 1.794, "limit": 2.0}; no_runs ok {"wet_max_um": 193.9, "travel_mm": 0.678, "limit_wet_um": 250.0}; overspray ok {"fraction": 0.0404, "mist_fraction": 0.2887, "limit": 0.05}; untouched ok {"area_cm2": 0.0, "limit_cm2": 0.5}



clear-coat: video contact sheet

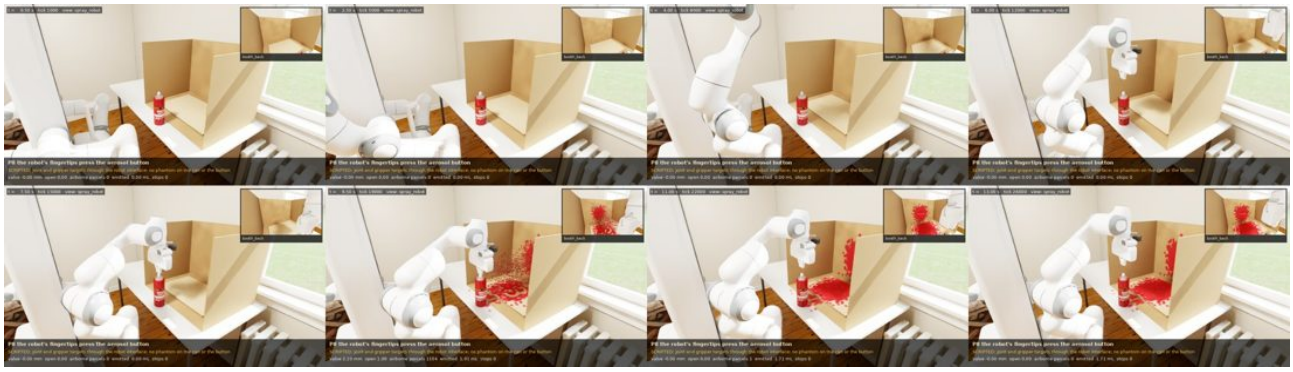


clear-coat: dry film after the episode (white outline: the task's target)

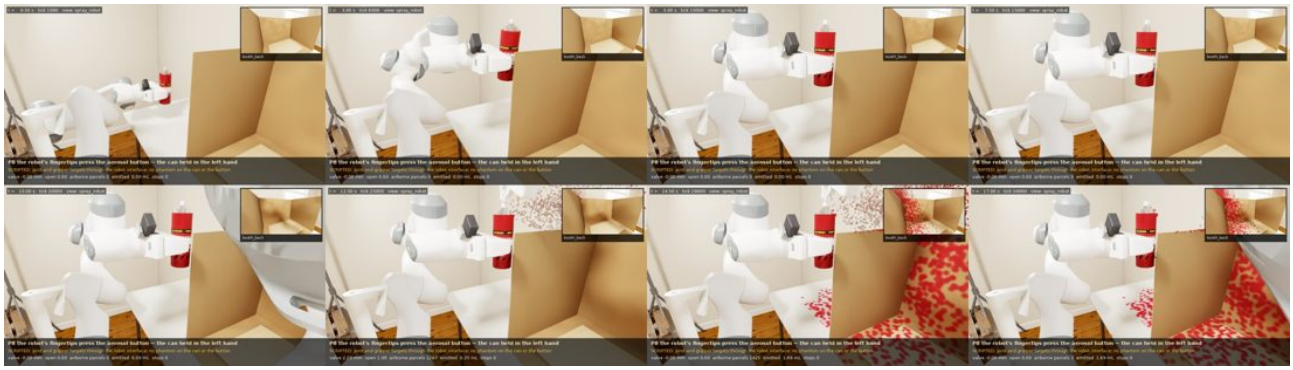
4. Robot demonstrations (scripted through the robot interface)



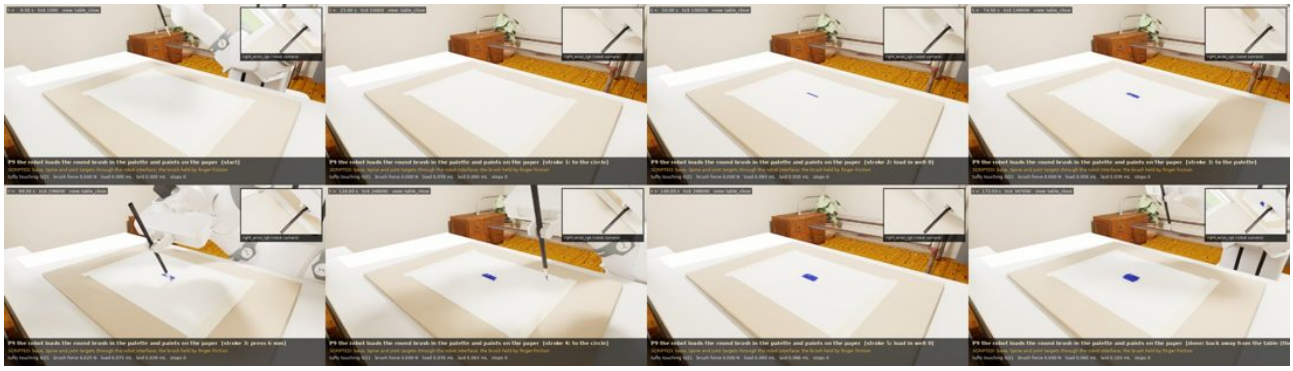
P6 the robot rolls the wall, roller held by finger friction



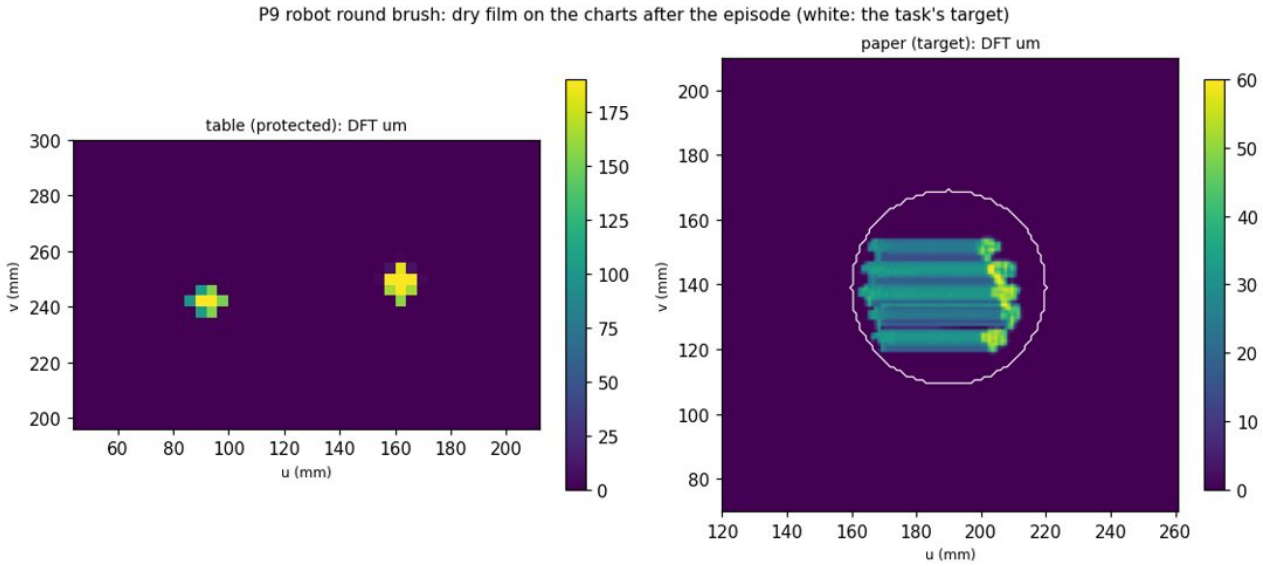
P8 the robot's fingertips press the aerosol button



P8 bimanual: the left hand holds the can at the spray tasks' start grasp, a right fingertip opens the valve

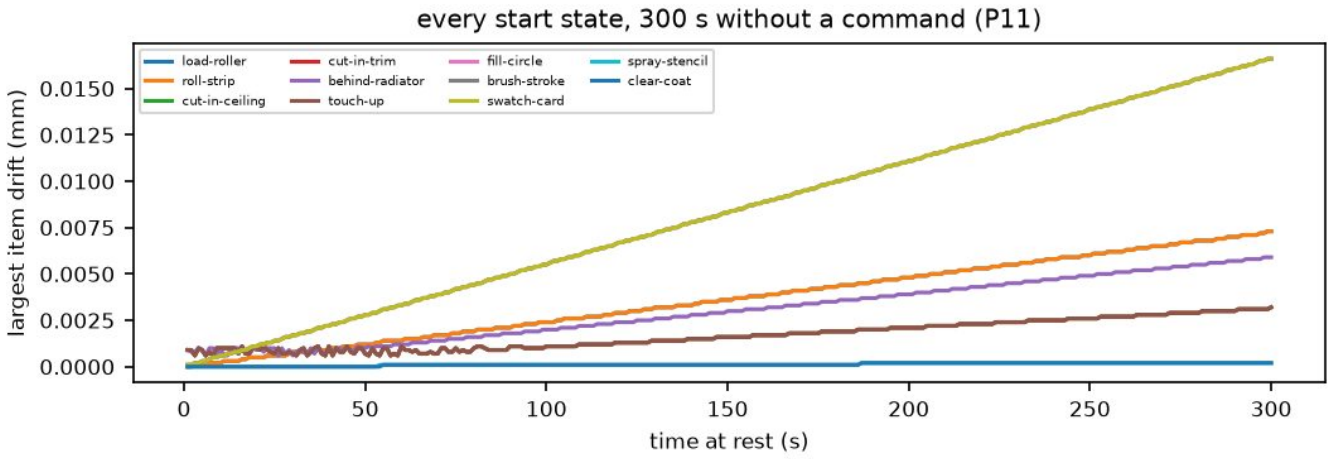


P9 the robot loads the round brush in the palette and paints the paper



P9 dry film on the paper after the robot's strokes

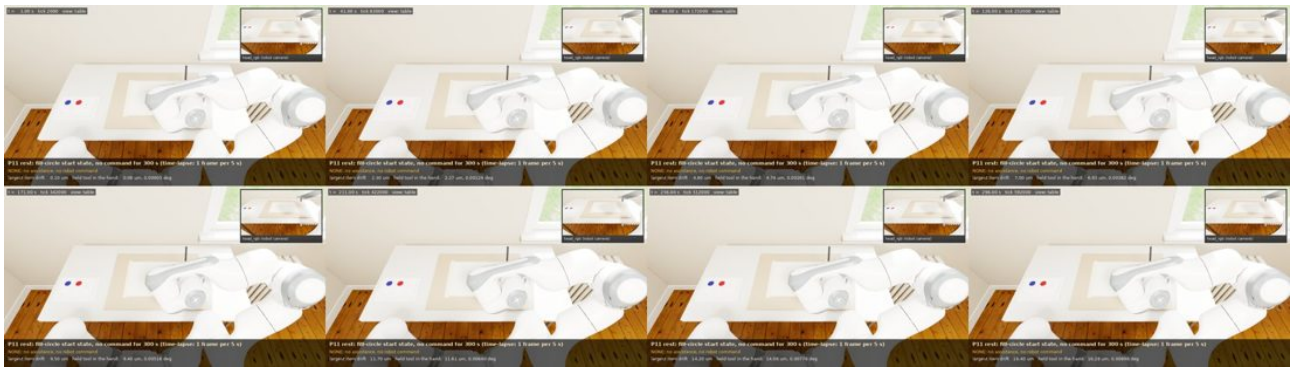
5. Stability, stress, layouts, evaluator, runtime



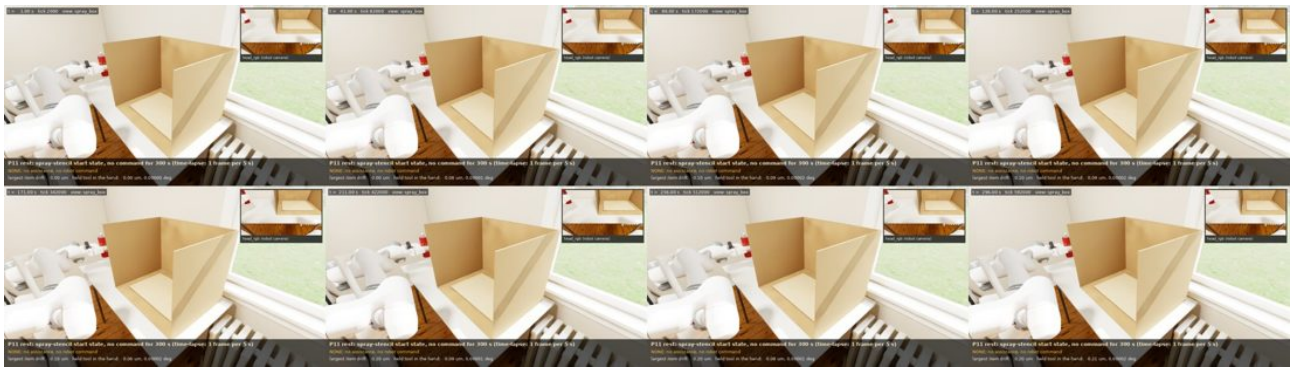
P11 rest: drift of every item over 300 s per task



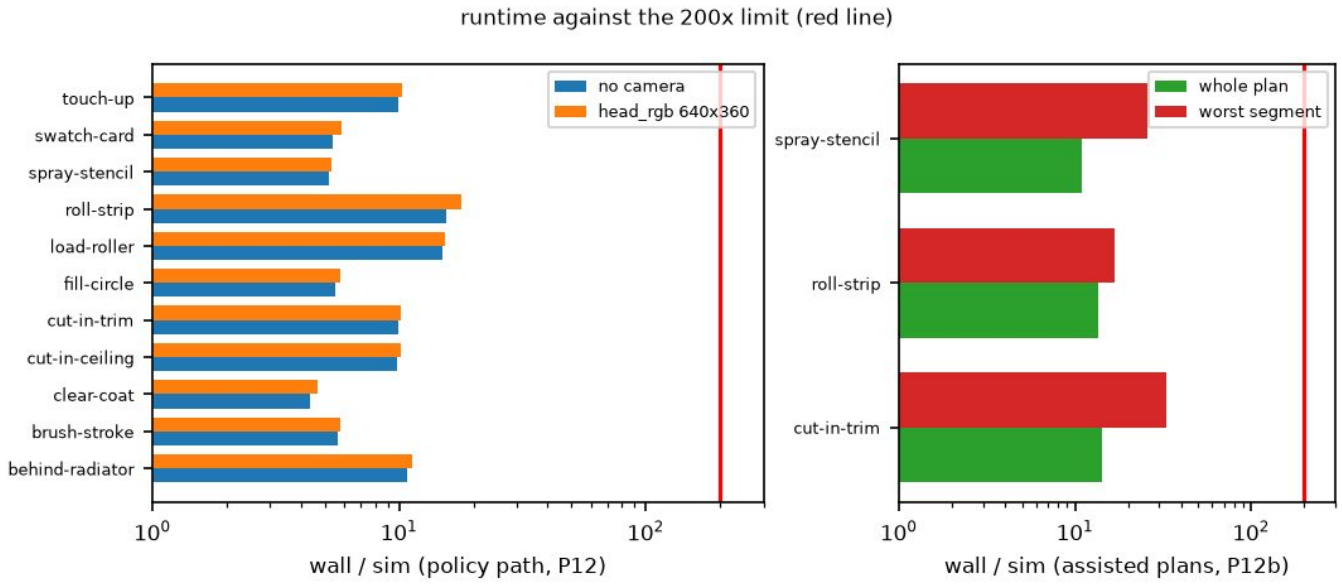
P11 rest time-lapse, roll-strip start (one frame per 5 s; the held tool's offset in the hand overlaid)



P11 rest time-lapse, fill-circle start (one frame per 5 s; the held tool's offset in the hand overlaid)



P11 rest time-lapse, spray-stencil start (one frame per 5 s; the held tool's offset in the hand overlaid)



P12 runtime per task, with and without cameras

task	case	accepted	failed	as expected
roll-strip	honest	False	roller_in_tray	True
roll-strip	thin	False	hide, roller_in_tray	True
roll-strip	heavy	False	no_runs, roller_in_tray	True
roll-strip	lap	False	no_laps, roller_in_tray	True
roll-strip	casing	False	protected, roller_in_tray	True
roll-strip	holiday	False	uniform, roller_in_tray	True
roll-strip	elbow	False	untouched, roller_in_tray	True
cut-in-ceiling	honest	True	-	True
cut-in-ceiling	short	False	reaches_corner	True
cut-in-ceiling	ceiling	False	protected	True
cut-in-trim	honest	True	-	True
cut-in-trim	casing_face	False	protected	True
behind-radiator	honest	True	-	True
behind-radiator	radiator	False	protected	True
touch-up	honest	True	-	True
touch-up	blob	False	no_ridge	True
touch-up	missed	False	scuffs_hidden	True
fill-circle	honest	True	-	True
fill-circle	overrun	False	within_line	True
swatch-card	honest	True	-	True
swatch-card	corner	False	coverage	True
brush-stroke	honest	True	-	True
brush-stroke	constant6mm	False	within_line	True
spray-stencil	honest	True	-	True
spray-stencil	heavy	False	coat, no_runs	True
spray-stencil	bleed	False	sharp_edges	True
clear-coat	honest	True	-	True
clear-coat	uneven	False	coat, even, no_runs	True
clear-coat	thin	False	coat	True

6. Parameter ledger

name	value	unit	label	basis
AIR	{"rho": 1.2, "mu": 1.81e-05}	kg/m3, Pa s	published	dry air at 20-23 C
CLIMATE	{"T_C": 23.0, "RH": 0.5}	C, -	authored	seeded 18-27 C and RH 30-70 % (variants)
WATER_RHO	997.0	kg/m3	published	water at 25 C
LATEX	{"rho": 1295.3, "phi0": 0.42, "sigma": 0.032, "eta_inf": 0.1796, "eta0_min": 2.258, "eta0_max": 256.6, "gamma_c": 0.0308	SI (S_um per um of dry film)	datasheet + derived + estimate	rho, phi0 [S1]; eta/tau_y fit M2 [S2, S3] (tau_y_max 2 Pa and t_r 20 s estimates); sigma 30-35 mN/m estimate; phi_m 0.64 random close packing, phi_we 0.570 = viscosity x10 (M2); k_m 0.006 m/s from Eastman wet edge 2-4 min [S9] (M2); S 0.187 /um is an optical estimate, NOT a product hiding measurement or a consequence of rated spread rate. Class sensitivity bracket 0.08-0.30 /um from ~2-8 S/mil for
LATEX_OPTICAL_RANGE	[0.08, 0.3]	1/um	estimate bounded by coating-class data	Dow [S60] figures 3/7 span roughly 2-8 S/mil in acrylic/TiO2 formulations (divide by 25.4); a sensitivity envelope, not a confidence interval or ProMar-specific drawdown. Nominal S is 0.187.
ACRYLIC	{"rho": 1350.0, "phi0": 0.45, "sigma": 0.035, "eta_inf": 0.3, "eta0_min": 1.0, "eta0_max": 8.0, "gamma_c": 1.0, "n": 0.4	SI	search + estimate	artist fluid acrylic: 800-1,200 cP Brookfield [S37] -> eta at low shear 1-8 Pa s, mild yield (estimates); semi-transparent artist pigments S 0.06 /um (estimate); drying and binder haze as latex (same water-borne binder; estimate; a haze of 0.6 would be three times the pigment's own scattering)
ENAMEL	{"rho": 1100.0, "phi0": 0.25, "sigma": 0.028, "eta_inf": 0.03, "eta0_min": 0.03, "eta0_max": 0.05, "gamma_c": 1.0, "n":	SI	datasheet + estimate	aerosol enamel after propellant flash (Painter's Touch class [S31]: 1-2 mil DFT, touch 20 min; acetone, n-butyl acetate, naphtha, xylene): Newtonian 0.03 Pa s at landing, solids 0.25, k_m x5 for the volatile solvent (touch in 20 min); its evaporation does not depend on the room's water humidity (solvent=True: the rate is taken at the reference 50 % RH, so the nominal climate is unchanged); all est
LACQUER	{"rho": 1000.0, "phi0": 0.2, "sigma": 0.028, "eta_inf": 0.03, "eta0_min": 0.03, "eta0_max": 0.05, "gamma_c": 1.0, "n": 1	SI	estimate	clear lacquer aerosol as the enamel without pigment (S = K = 0)
TINT_BASES	{"white": [0.7, 1.0], "pastel": [0.4, 0.58], "medium": [0.15, 0.33], "deep": [0.0, 0.02]}	LRV (Y) minimum, scattering / white base	search + calibrated + estimate	tint bases by TiO2 share of the solids: white ~30 %, pastel 15-20 %, medium 8-12 % [S49] -> scattering in proportion (17.5/30, 10/30; estimates). Deep/accent bases carry no TiO2 (deep-tone bases are the pastel formula with the TiO2 removed [S54]), so their scattering is the extenders' and colorants': the factor 0.020 is calibrated on coats-to-hide for a deep colour (SW 6244 Naval): one 43 um coat
COLORS	{"alabaster": [237, 234, 224], "extra_white": [238, 239, 234], "lullaby": [203, 212, 212], "accessible_beige": [209, 199	sRGB 0-255	search / authored	Sherwin-Williams SW 7008 Alabaster, SW 7006 Extra White, SW 9136 Lullaby (sRGB 203/212/212), SW 7036 Accessible Beige, SW 6244 Naval [S46] (search-level); a mid-grey P-Shade tinted primer under a deep colour (ProMar 200 TDS: 'use the recommended P-Shade primer to optimize hide' [S1]; the shade for Naval is an estimate); artist and substrate colours authored
FILM	{"every_ticks": 40, "max_substeps": 512, "cfl": 0.4, "h_min": 1e-09, "set_thin": 2e-07, "struct_after_shear": 0.0, "tex_	ticks, -, -, m, m, -, m	derived / estimate	film transport every 20 ms with CFL substeps (M2: explicit limits 0.35-5.6 s at 1-2 mm cells after application); a film thinner than 0.2 um sets at once (estimate); an applicator's passage leaves lambda = 0 (steady-state structure at 1e4/s, M2); sub-grid texture wavelengths: roller stipple 2 mm, brush marks 1 mm, spray orange peel 0.6 mm (estimates within the 0.5-5 mm range of M2's levelling table
CELL	{"wall": 0.002, "trim": 0.002, "ceiling": 0.002, "floor": 0.004, "tray": 0.002, "paper": 0.001, "crate": 0.001, "stencil	m	derived	M2: 2 mm resolves roller stipple and 1 mm brush marks and stencil edges; P2 levelling needs >= 8 cells per wavelength; the task metrics at 1 mm against 2 mm: the E7 and P3 refinement runs
SUBSTRATE	{"primed_drywall": {"sorptivity": 1e-07, "capacity": 5e-06}, "spackle": {"sorptivity": 1e-06, "capacity": 3e-05}, "water	m/s^0.5, m	datasheet + estimate	watercolour paper: Cobb60 23 g/m2 (300 g/m2 cotton [S36]) -> S = 23e-6 m / sqrt(60 s); capacity 0.51 mm x 40 % pores (estimate); PVA-primed drywall 0.1 um/s^0.5 and spackle x10 (estimates); canvas drop cloth, cardboard: estimates
ROLLER	{"cover_len": 0.2286, "core_od": 0.0424, "nap": 0.0095, "n_around": 72, "n_along": 46, "E_nap": 40000.0, "beta": 3.0, "c	m, -, Pa, -, Pa s/m, -, m/s, -, -, -, 1/s, -, N m, kg	datasheet + derived + estimate	9 in x 3/8 in cover on a 1.5 in core, OD 61 mm [S20] (M5); nap modulus 40 kPa (20-80, estimate; no compression curve found), densification beta 3 (estimate), nap damping 2e3 Pa s/m (its hysteresis is the rolling resistance, ~5-10 % of the normal force at 0.3-0.5 m/s; estimate); nap-wall friction 0.6 dry / 0.4 wet (estimates); regularisation 20 mm/s (stability: c dt / m_eff < 1); pore fraction 0.9
TRAY	{"size": [0.368, 0.279, 0.089], "well_len": 0.12, "well_depth": 0.08, "ramp_drop": 0.07, "wall": 0.0025, "mass": 0.25, "	m, kg	retailer + estimate	Wooster R404 deep-well tray 14.5 x 11 x 3.5 in, 2 qt [S22]; well/ramp split, rib height and 0.25 kg PP mass are estimates
PRBM	{"gamma": 0.85, "k_theta": 2.65}	-	published	Howell, Compliant Mechanisms (2001): pseudo-rigid-body cantilever, characteristic radius 0.85 and stiffness coefficient 2.65 (literature)
BRUSHES	{"sash": {"width": 0.0635, "thick": 0.0159, "length": 0.0746, "bevel_deg": 15.0, "tufts": [24, 6], "filaments": 6642, "E	m, Pa, kg, N/m	datasheet + derived + estimate	Purdy XL Glide 2.5 in angled sash: 63.5 x 15.9 mm, 74.6 mm out [S12], Tynex 612 wet modulus 2.9 GPa [S13], long-reach dog-leg brush for behind radiators (ProDec PDLB50 class: 2 in head set at 35 deg to a wooden handle, 355 mm overall, 60 mm bristles [S50]; Hamilton radiator brush 0.12 kg [S50]; thickness and filament count by the sash brush's packing, estimates), 12/8 mil taper (DuPont standard ta
BRUSH_PAINT	{"k_up": 2.0, "hold": 0.75, "residual": 0.25, "k_drip": 0.5, "c_tip": 0.05, "D_bundle": 0.0001, "h_max": 0.00015}	1/s, -, -, 1/s, N s/m, m2/s	published + calibrated + estimate	paint in the bristle bundle: the full free length's pore volume (packing, M4) wetted up to the dip depth at k_up (Washburn in ~40 um gaps: seconds; estimate); drains by drops (k_drip, estimate) to 'hold' of the wetted pore volume after withdrawal (ASTM D5068 hang [S14]); a residual saturation of 0.25 of the wetted pores never leaves (capillary hold-up: 1 - 0.25/0.75 = 67 % of a drained load is rel
PAIL	{"r_top": 0.089, "r_bottom": 0.08, "height": 0.14, "wall": 0.0015, "mass": 0.14}	m, kg	retailer + estimate	Bercom Handy Paint Pail 1 qt: 7 in x 5.5 in tall [S23] (taper estimate)
CAN	{"r": 0.0838, "height": 0.19, "wall": 0.0004, "mass": 0.35, "fill": 0.0025}	m, kg, m3	estimate	US 1-gal steel paint can 6-5/8 in x 7.5 in (common size, estimate); 2.5 L left in it
AEROSOL	{"r": 0.033, "height": 0.196, "can_mass": 0.12, "fill_mass": 0.34, "button_r": 0.0075, "button_h": 0.012, "stroke": 0.00	m, kg, N, N/m, kg/s, m/s, deg, m, -	search + published + calibrated + estimate	12 oz can 66 x 196 mm (estimate); valve stroke 1.5-2.5 mm and finger force 15-30 N, spring 8-25 N [S29] -> preload 10 N + 7.5 N/mm (full open 23.5 N at 1.8 mm); discharge 2.0 g/s: the high-output 'broad-stream' class (15-35 g / 10 s, Kwasny US 5957341 [S29]; conventional valves ~10 g / 10 s) that a '2X' can [S31] is (estimate inside the range; a 12 oz can empties in ~3 min); launch 25 m/s, 12 deg

name	value	unit	label	basis
STENCIL	{"size": [0.4, 0.3], "thick": 0.0003, "rho": 1390.0, "mu": 0.35, "seal_tolerance": 2e-05}	m, kg/m3	search + estimate	12 mil (0.30 mm) clear plastic stencil [S26] cut from a 16 x 12 in blank (blanks run 8.5 x 11 to 12 x 18 in; a 30 x 20 cm sheet leaves only 1 cm above and below the 18 cm motif, inside the spray's halo), PET density 1,390 kg/m3; rigid plate on a flat sealed panel. Ideal smooth-contact seal only within 20 um local clearance (authored numerical tolerance, not measured roughness/adhesion); resolved o
ROOM	{"x": 3.66, "y": 3.35, "z": 2.438, "wall_t": 0.1}	m	authored	US 11 x 12 ft bedroom, 8 ft ceiling
WINDOW	{"x": 0.55, "w": 0.914, "sill": 0.813, "head": 2.032, "casing": 0.057, "casing_t": 0.0175, "stool_d": 0.075, "stool_t":	m	search + authored	36 x 48 in opening, 2-1/4 in colonial casing 11/16 in proud [S44]; stool and apron dimensions authored
BASEBOARD	{"h": 0.083, "t": 0.0127}	m	search	3-1/4 in x 1/2 in [S44]
OUTLET	{"x": -0.9, "z": 0.356, "w": 0.07, "h": 0.114, "t": 0.005}	m	search	duplex plate 2.75 x 4.5 in, 14 in to centre [S44]
RADIATOR	{"len": 0.64, "depth": 0.229, "height": 0.635, "gap": 0.064, "sections": 10, "pipe_r": 0.0167, "pipe_x": 0.22}	m	authored	4-column cast-iron radiator, 25 in tall, back 2.5 in off the wall, 10 sections at the 2.5 in (63.5 mm) pitch of 4-column cast iron (typical, authored)
TABLE	{"size": [1.22, 0.61], "height": 0.74, "top_t": 0.025, "mass": 10.0, "leg_r": 0.0127}	m, kg	authored	Lifetime-class 4 ft folding table (estimate)
EVAL	{"hide_dft": 3.3e-05, "hide_dE": 1.0, "hide_frac": 0.98, "run_h": 0.00025, "sag_move": 0.001, "lap_area": 0.0001, "dE":	m, -, m, m, m2, -, m2, s	derived + authored	hide: the dry stack within dE*ab 1.0 of the paint's own infinitely thick colour over the old colour under it (authored perceptual criterion; optical coefficients are estimates with the coating-class sensitivity range of the optical sensitivity check, not a product-backed hiding-power measurement; the DFT is reported); runs: wet film > 250 um (multinotch sag range M2/[S45]) before the settle (progr

7. References

The source keys cited in this report.

[S1] Sherwin-Williams, ProMar 200 Zero VOC Interior Latex Eg-Shel B20 series product data sheet (coverage 350-400 ft2/gal at 4 mils wet, 1.7 dry; 42 +- 2 % volume solids; 10.81 lb/gal; touch 1 h, recoat 4 h; 15-20 GU at 85 deg). sherlink.sherwin.com / paintdocs.com PDS 101.83A.

[S2] TA Instruments, "Rheological Characterization of Paints and Coatings", AZoM article 12120 (Table 1 profile of a thin-film waterborne paint; HSV 50-150 mPa s; LSV at 0.3 1/s; levelling 25-100 Pa s, sag below 10 Pa s).

[S3] R. D. Hester, D. R. Squire, "Rheology of Waterborne Coatings", J. Coatings Technology 69 (864), Jan 1997, paint.org PDF (yield/viscosity before and after shear; levelling stress; viscosity during drying).

[S9] Eastman Chemical, "Development of Low VOC Additives to Extend the Wet Edge and Open Time of Aqueous Coatings" (open time 5 to 7 min, wet edge 2 to 4 min, 3-mil Bird).

[S12] Purdy XL Glide trim brushes product page (2.5 in: 5/8 in thick, 2-15/16 in length out; Tynex nylon / Orel polyester) and retailer listings.

[S13] DuPont/Brushtec, "Filament Performance in Brushes" brochure (Table 1 moduli: Tynex 612 580/420 kpsi at 50 % RH / wet, hog bristle 680/410); DuPont Filaments, "How to Evaluate a Paintbrush" (standard tapers 15/10, 12/8 mil; brush deflection test).

[S14] ASTM D5068-04(2013) Standard Practice for Preparation of Paint Brushes for Evaluation (dip to half the filament length, 10 s, drain 30 s, weigh).

[S20] Wooster 9 in x 3/8 in roller covers (Lowe's, HD Supply listings: 1.5 in core).

[S22] Wooster R404 Deep-Well tray (PaintSupply: 14.5 x 11 x 3.5 in, 2 qt).

[S23] Bercom Handy Paint Pail 2500 (1 qt; 7 x 5.5 in; magnetic brush holder, scraper lip).

[S26] Cutting Edge Stencils product pages (12 mil clear plastic vs 5-8 mil mylar); 10 mil mylar listing.

[S29] US 8528837 (aerosol discharge 10 g/10 s, up to 35 g/10 s); aerosolmeteredvalve.com valve guide (spring 8-25 N, finger 15-30 N, stroke 1.5-2.5 mm, 3.5-5.2 bar abs); globalaerosols.com (1.5-2.5 kgf); US 7686193 (trigger 3:1).

[S31] Rust-Oleum Painter's Touch 2X Ultra Cover technical data sheets (12 oz; 1-2 mil DFT; 8-12 ft2/can with 15 % loss; 10-16 in; touch 20 min).

[S36] Reflex Aquarell Da Capo data sheet (300 g/m2, 510 um, Cobb60 23 g/m2); James Cropper Kendal watercolour board (190 g/m2, Cobb 40 g/m2); ISO 535 Cobb method; ForMatter watercolour paper notes.

[S37] Golden Artist Colors technical charts (Heavy Body 23,000-27,000 cP; Fluid Acrylics 800-1,200 cP).

[S44] Outlet height 12-16 in to centre; baseboard 3-5.25 in; casing 2.25-3.5 in (my-bliss-home, dgfloors, electricalflux, engineerfix).

[S45] ASTM D4400 sag resistance (multinotch applicator; Leneta anti-sag meter 3-12 mil notches, wet film about half the clearance).

[S46] Sherwin-Williams SW 9136 Lullaby (LRV 64.67, sRGB 203/212/212) listings; SW 7036, SW 7008, SW 6244 values.

[S49] Tint bases by TiO2 share of the solids: white ~30 %, pastel 15-20 %, medium 8-12 %, deep/accent 3-5 % (or none) (usa-painter.org FAQ "What is tinted paint"; cypaint.com "Base A and D" and "Behr bases"; woodworkingadvisor "Deep base vs white base").

[S50] Long-reach dog-leg brushes for behind radiators: ProDec PDLB50 2 in dog-leg, head at 35 deg, 355 mm overall, 60 mm bristles; ProDec NC2 2 in, 300 mm handle; Harris 2 in long-reach; Hamilton extra-long radiator brush 0.12 kg (retailer listings). Radiator rollers: ProDec 21 in (525 mm) and 24 in long-arm mini-roller frames.

[S54] "On the Rheology of Tint Bases", CoatingsTech / paint.org (2013): pastel and deep-tone tint bases; the key difference is the removal of TiO2 in the deep-tone formulations.

[S60] Dow, "Reducing Titanium Dioxide in Paints for a More Sustainable Future", figures 3 and 7.
https://www.dow.com/content/dam/dcc/documents/926/926-02459-01-reducing-titanium-dioxide-in-paints-for-a-more-sustainable-future.pdf